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# THE LOCAL NEIGHBORLINESS THRESHOLD OF THE SYMMETRIC MOMENT CURVE: AN INTEGRAL CRITERION, SHARP ASYMPTOTICS, AND THE BARVINOK–LEE–NOVIK CONJECTURE FOR $k \le 440$ AND FOR ALL SUFFICIENTLY LARGE $k$

PAATA IVANISVILI

ABSTRACT. Let $\phi_k$ be the largest $\phi$ such that any $k$ points of the symmetric moment curve $U_k(t) = (\cos t, \sin t, \cos 3t, \sin 3t, \dots, \cos(2k-1)t, \sin(2k-1)t)$ lying on an open arc of length $\phi$ span a face of $\operatorname{conv} U_k$. Barvinok, Lee and Novik proved $\pi/2 < \phi_k < \pi$ and $\phi_k \to \pi/2$, reduced $\phi_k$ to configurations collapsed to the two endpoints of the arc, and conjectured that the balanced collapse is always the worst one. We turn their characterization into an explicit two-integral criterion. We use it to prove: $\phi_k < \frac{\pi}{2} + \frac{\log(8\pi(k+1/2))}{4(k-1)}$ for even $k \ge 4$; $\phi_k < \frac{\pi}{2} + \frac{\log(64\sqrt{\pi}(k-1)\sqrt{k-1/2})}{2k-8}$ for odd $k \ge 13$; and $\liminf_k \left(4k(\phi_k - \pi/2) - \log k\right) \ge 2\log 2$. We also prove the effective two-sided bound $\frac{\pi}{2} + \frac{\log k}{4k} < \phi_k < \frac{\pi}{2} + \frac{\log k + 4.1}{4k}$ for every $k \ge 2$; on a finite range of $k$ it rests on explicit closed-form inequalities certified in ball arithmetic. Refining the analysis we obtain the exact second-order term, $\lim_k \left(4k(\phi_k - \frac{\pi}{2}) - \log k\right) = \log(8\pi)$, which is the asymptotics of the balanced split. We prove the Barvinok–Lee–Novik conjecture for all $k \le 440$ (both parities; computer-assisted, ball arithmetic), together with the monotonicity of $\phi_k$ in that range. We also prove it, analytically but with an ineffective threshold, for all sufficiently large $k$. The splits far from balanced are excluded by the sharp asymptotics. For near-balanced splits we show that the criterion is strictly concave in the exponent: its second derivative is at most $-4\delta/N + O(N^{-2})$ with $N\delta \asymp \log k$. This is proved with Stein identities for the weights $\sin^a u \sin^b(\ell - u)$ and a moving-frame treatment of the kink $|\sin(u-s)|$. The same estimates, used to first order in the other exponent, show that $\phi_k$ is strictly decreasing for all sufficiently large $k$. This answers BLN's monotonicity question asymptotically. The conjecture thus remains open only on a finite but unspecified range $440 < k < K_0$. We also isolate an "endpoint monotonicity" conjecture which implies $\phi_{k+1} < \phi_k$.

## 1. INTRODUCTION

Let $\mathbb{S} = \mathbb{R}/2\pi\mathbb{Z}$ and $B_k = \operatorname{conv} U_k(\mathbb{S}) \subset \mathbb{R}^{2k}$, the Barvinok–Novik orbitope [1, 3]. A *raked trigonometric polynomial* of degree $\le 2k-1$ is $f(t) = c + \sum_{j=1}^{k}(a_j \cos(2j-1)t + b_j \sin(2j-1)t)$, i.e. $f = c + \langle C, U_k\rangle$. Faces of $B_k$ through $U(t_1), \dots, U(t_n)$ correspond to nonnegative raked polynomials vanishing at the $t_i$.

**Definition 1** ([2])**.** $\phi_k$ is the largest number such that for every open arc $\Gamma \subset \mathbb{S}$ of length $\phi_k$ and all distinct $t_1, \dots, t_n \in \Gamma$, $n \le k$, the set $\operatorname{conv}\{U(t_1), \dots, U(t_n)\}$ is a face of $B_k$.

**Known.** Barvinok–Novik [1]: $\phi_k > 0$. Lee: $\phi_k > \sqrt{6}\, k^{-3/2}$. Barvinok–Lee–Novik [2]: $\pi/2 < \phi_k < \pi$ and $\phi_k \to \pi/2$; $\phi_2 = 2\pi/3$, $\phi_3 = \pi - \arccos\frac{3-\sqrt{5}}{2}$, and $\phi_4 \approx 1.8706585$ in closed form. They also give a characterization (Theorem 13 below). Edges of $B_k$ were determined in [1, 4]. The facial structure of $B_3$ is unknown.

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**Conjecture 2** ([2]). *In Theorem 13(2) one can take $m_a = m_b = k$ if $k$ is even and $\{m_a, m_b\} = \{k-1, k+1\}$ if $k$ is odd. For even $k$ this is equivalent to $\phi_k = 2\alpha_k$, where $\alpha_k$ is the smallest positive root of $\cos\alpha + 1 + \sum_{j=1}^{k-1}(-1)^j \frac{(2j-1)!!}{(2j)!!}\tan^{2j}\alpha = 0$.*

BLN also write that their computations suggest $\phi_k$ is decreasing, but they could not prove it.

**Results.** For a *split* $(p,q)$, $p,q \geq 1$, $p+q=k$, and $L \in (0,\pi)$, let $f_{p,q,L}$ be the unique raked polynomial of degree $\leq 2k-1$ with constant term 1, a root of multiplicity $2p$ at 0 and of multiplicity $2q$ at $L$ [2, Cor. 3.5]. Let $L(p,q) \in (0,\pi]$ be the first $L$ at which $f_{p,q,L}$ acquires an additional root ($L(p,q) = \pi$ if none). Put $a = 2p-1$, $b = 2q-1$.

**Theorem 3** (integral criterion). *For $L \in (0,\pi)$ let $A(u) = \sin^a u \sin^b(L-u)$ on $(0,L)$, let $s \in (0,L)$ be the argument of $\int_0^L A(u)e^{iu}du$, and put*

$$I_{\text{in}} = \int_0^s A(u)\sin(s-u)\,du = \tfrac{1}{2}\int_0^L A(u)|\sin(u-s)|\,du, \qquad I_{\text{out}} = \int_L^\pi \sin^a u \sin^b(u-L)\sin(u-s)\,du.$$

*Then $f_{p,q,L} \geq 0$ on $\mathbb{S}$ iff $I_{\text{out}} \geq I_{\text{in}}$, and $f_{p,q,L}$ is $\geq 0$ and has a root outside $\{0,L\}$ iff $I_{\text{out}} = I_{\text{in}}$. Consequently, $f_{p,q,L}$ has a root outside $\{0,L\}$ iff $I_{\text{out}} \leq I_{\text{in}}$. Moreover $\Delta_{p,q}(L) := I_{\text{out}} - I_{\text{in}}$ is positive on $(0, L(p,q))$, vanishes at $L(p,q)$ if $L(p,q) < \pi$, and is negative on $(L(p,q),\pi)$. Finally $\phi_k = \min_{p+q=k} L(p,q)$ and $L(p,q) = L(q,p)$.*

**Theorem 4** (upper bounds). *(i) For even $k \geq 4$: $\phi_k \leq L(\frac{k}{2},\frac{k}{2}) < \dfrac{\pi}{2} + \dfrac{\log\left(8\pi(k+\frac{1}{2})\right)}{4(k-1)}$.*

*(ii) For odd $k \geq 13$: $\phi_k < \dfrac{\pi}{2} + \dfrac{\log\left(64\sqrt{\pi}\,(k-1)\sqrt{k-1/2}\right)}{2k-8} = \dfrac{\pi}{2} + \dfrac{3\log k + O(1)}{4k}$.*

**Theorem 5** (lower bounds). *(i) $\liminf_{k\to\infty}\left(4k(\phi_k - \frac{\pi}{2}) - \log k\right) \geq 2\log 2$.*
*(ii) $\phi_k > \frac{\pi}{2} + \frac{\log k}{4k}$ for every $k \geq 2$.*

The proof of (ii) is analytic for $k > 2\cdot 10^4$. For $301 \leq k \leq 2\cdot 10^4$ it consists of certifying in ball arithmetic, for each of the $\approx 10^8$ splits, the sign of an explicit closed form in Gamma functions (Proposition 25). For $k \leq 300$ it follows from the enclosures of Theorem 8.

**Theorem 6** (effective upper bound). $\phi_k < \frac{\pi}{2} + \frac{\log k + 4.1}{4k}$ *for every $k \geq 2$.*

For even $k \geq 302$ this follows from Theorem 4(i), and for $k \leq 300$ from Theorem 8. For odd $k \geq 301$ it comes from an explicit criterion for the split $(\frac{k-1}{2},\frac{k+1}{2})$ (Proposition 19). That criterion is certified in ball arithmetic for $k \leq 200001$ and verified by hand beyond. Together with Theorem 5(ii),

$$\frac{\pi}{2} + \frac{\log k}{4k} < \phi_k < \frac{\pi}{2} + \frac{\log k + 4.1}{4k} \qquad \text{for every } k \geq 2.$$

**Theorem 7** (sharp asymptotics). $\lim\limits_{k\to\infty}\left(4k\big(\phi_k - \tfrac{\pi}{2}\big) - \log k\right) = \log(8\pi)$, *i.e.* $\phi_k = \frac{\pi}{2} + \frac{\log k + \log(8\pi)}{4k} + o\big(\frac{1}{k}\big)$.

This is exactly the asymptotics of the balanced split. It is what Conjecture 2 predicts, but it does not imply the conjecture: a non-balanced split could still win at order $o(1/k)$. The proof (§6) is analytic but does not give an explicit rate. The cruder bounds of Theorems 4, 5(ii) and 6 hold for every $k$ in their stated ranges; Theorem 5(i) is an asymptotic statement.

**Theorem 8** (computer-assisted, rigorous). *For every $2 \leq k \leq 440$, the balanced split is the unique minimizer in Theorem 13(2). That is, Conjecture 2 holds, in both parities, and $\phi_k = L(\lfloor k/2\rfloor, \lceil k/2\rceil)$. Moreover $\phi_{k+1} < \phi_k$ for $2 \leq k < 440$, and $L(p,q+1) < L(p,q)$ for all $p+q <$*

200. *Each $\phi_k$ is enclosed in an interval of half-width $1.01 \cdot 10^{-12}$ about a floating-point value, printed to 12 decimals (files* `numerics/rigorous_200.txt`, `numerics/rigorous_201_300.txt`, `numerics/rigorous_301_400.txt` *and* `numerics/rigorous_401_440.txt`*).*

BLN proved $k \le 4$. The proof is in §8: Theorem 3 turns each comparison into the sign of an explicit closed-form expression at a rational point, and these signs are certified in ball arithmetic (Arb).

**Theorem 9** (all sufficiently large $k$)**.** *There is $K_0$ such that for every $k \ge K_0$ the balanced split is the unique minimizer in Theorem 13(2), up to $p \leftrightarrow q$. That is, Conjecture 2 holds for all $k \ge K_0$.*

**Theorem 10** (monotonicity for large $k$)**.** *There is $K_2$ such that $\phi_{k+1} < \phi_k$ for all $k \ge K_2$. Moreover, endpoint monotonicity $L(p, q+1) < L(p, q)$ (Conjecture 11 below) holds for all $k = p + q \ge K_2$ with $0.42k \le p \le 0.58k$.*

The proofs (§7) are analytic but ineffective: we do not compute $K_0$ or $K_2$. Together with Theorem 8, the conjecture holds for $k \le 440$ and for $k \ge K_0$. The splits far from balanced are handled by the sharp asymptotics. For near-balanced splits we prove that the criterion $\log I_{\text{in}} - \log I_{\text{out}}$ is strictly concave in the (real) exponent $2p - 1$ at $L = L(\lfloor \frac{k}{2} \rfloor, \lceil \frac{k}{2} \rceil)$. Its second derivative is at most $-4\delta/N + O(N^{-2})$ with $N\delta \asymp \log k$.

In floating point (§9) we also find, for $k \le 300$, that $p \mapsto L(p, k-p)$ is strictly decreasing on $p \le k/2$, and that $4k(\phi_k - \pi/2) - \log k$ approaches $\log(8\pi) = 3.2242$, the value of the upper bound in Theorem 4(i). So the upper bound (i) is asymptotically sharp if the conjecture holds. We also propose:

**Conjecture 11** (endpoint monotonicity)**.** *$L(p, q+1) < L(p, q)$ for all $p, q \ge 1$.*

Conjecture 11 implies $\phi_{k+1} \le L(p_*, q_* + 1) < L(p_*, q_*) = \phi_k$, where $(p_*, q_*)$ attains $\phi_k$. So it answers BLN's monotonicity question. It is rigorously verified for $p + q < 200$ (Theorem 8).

*Relation to the source paper.* On an arc of length $< \pi$ the curve $U_k$ meets every hyperplane at most $2k$ times, counting multiplicity [2, Thm. 3.1]; that is, it is a convex curve. The configurations in Theorem 3 are the endpoint principal representations, the analogue in even dimension of the upper and lower hulls in the source paper's Theorem 2.1. $\phi_k$ is the first time the corresponding supporting hyperplane of the convex hull of the *arc* stops supporting the whole closed curve. The criterion follows the source paper's method: factor the derivative of the support function into explicit sines, and read off signs from integrals of positive functions.

## 2. Facts from [2]

**Theorem 12** ([2], Thm. 3.1, Cor. 3.5)**.** *(a) A nonzero raked $f$ of degree $\le 2k-1$ has at most $4k-2$ roots on $\mathbb{S}$ counting multiplicity. If its constant term is 0, then $f(t+\pi) = -f(t)$, and it has at most $2k-1$ roots in any open semicircle. (b) If distinct $t_i$ lie in an open semicircle and $\sum m_i = 2k$, there is a unique raked $f$ of degree $\le 2k-1$ with constant term 1 having a root of multiplicity exactly $m_i$ at $t_i$.*

**Theorem 13** ([2], Thm. 5.1, Lemma 5.7)**.** *(1) For arcs of length $< \phi_k$: if $t_1, \dots, t_n \in \Gamma$ are distinct and $m_i$ are positive even integers with $\sum m_i = 2k$, then the polynomial of Theorem 12(b) has no other roots and is $\ge 0$. $\phi_k$ is the largest length with this property. (2) For an arc of length $\phi_k$ with endpoints $a, b$ there are positive even $m_a + m_b = 2k$ such that the polynomial with roots of multiplicities $m_a, m_b$ at $a, b$ is $\ge 0$ and has a root in $\Gamma + \pi$. (3) Let $\Gamma$ be an open arc of length $< \pi$ with endpoint $a$, and fix positive even $m_a, m_b$ with $m_a + m_b = 2k$. For $b \in \Gamma$ let $f_b$ be the polynomial of Theorem 12(b) with roots of multiplicities $m_a$ at $a$ and $m_b$ at $b$. Let $x, y, z \in \Gamma$, with $x$ between $a$ and $y$ and $y$ between $a$ and $z$. If $f_y \ge 0$ and $f_y$ has a root in $\Gamma + \pi$, then $f_x > 0$ off $\{a, x\}$, and $f_z$ is negative somewhere.*

## 3. The two-cluster criterion: proof of Theorem 3

**Lemma 14.** *Let $g$ be a raked polynomial of degree $\le 2k-1$ with zero constant term, i.e. an odd-harmonic trigonometric polynomial. If $g$ has roots of multiplicity $\ge \mu_j$ at points $r_j$ that are distinct mod $\pi$, with $\sum \mu_j = 2k-1$, then $g(t) = C\prod_j \sin^{\mu_j}(t-r_j)$.*

*Proof.* $\cos((2j-1)t)$ and $\sin((2j-1)t)$ are homogeneous polynomials of degree $2j-1$ in $(\cos t, \sin t)$. Multiplying by $(\cos^2 t + \sin^2 t)^{k-j}$, odd-harmonic polynomials of degree $\le 2k-1$ are exactly the restrictions $F(\cos t, \sin t)$ of binary forms $F$ of degree $2k-1$. A root of order $\mu$ at $t = r$ means the linear form $\ell_r(x,y) = x\sin r - y\cos r$ divides $F$ to order $\mu$, and $\ell_r(\cos t, \sin t) = -\sin(t-r)$. Distinct $r_j$ mod $\pi$ give non-proportional linear forms. Hence $\prod \ell_{r_j}^{\mu_j}$, of degree $2k-1$, divides $F$, and $F$ is a scalar multiple of it. □

*Proof of Theorem 3.* Write $f = f_{p,q,L}$. Then $f'$ is odd-harmonic of degree $\le 2k-1$, with roots of multiplicity exactly $2p-1$ at 0 and $2q-1$ at $L$ (Theorem 12(b)). Its antipodes are roots as well, since $f'(t+\pi) = -f'(t)$. By Rolle's theorem there is a further root $\sigma \in (0, L)$. Lemma 14 gives $f'(t) = K\, g(t)$ with $g(t) = \sin^a t \sin^b(t-L)\sin(t-\sigma)$. Since $f(0) = f(L) = 0$, $\int_0^L g = 0$. On $(0, L)$, $g = -A(u)\sin(u-\sigma)$ ($b$ is odd), so $\operatorname{Im}\left(e^{-i\sigma}\int_0^L Ae^{iu}\right) = 0$. The point $\int_0^L Ae^{iu}du / \int_0^L A$ lies in the closed circular segment bounded by the arc $\{e^{iu} : 0 \le u \le L\}$ and its chord. This is a convex set not containing 0, since $L < \pi$. The point is not one of the extreme points 1, $e^{iL}$, because $A > 0$ on $(0, L)$. The only points of the segment with argument 0 or $L$ are 1 and $e^{iL}$, so $s \in (0, L)$. The condition above says $\sigma \equiv s \pmod{\pi}$, and since $\sigma, s \in (0, L)$ with $L < \pi$, $\sigma = s$. Put $F(t) = \int_0^t g$. On $(0, s)$, $g > 0$; on $(s, L)$, $g < 0$; on $(L, \pi)$, $g > 0$, since every factor is positive there ($u - s \in (0, \pi)$). Hence $F > 0$ on $(0, L) \cup (L, \pi]$ and $F(\pi) = \int_L^\pi g = I_{\text{out}} > 0$. Since $g(t+\pi) = -g(t)$, we get $F(t+\pi) = F(\pi) - F(t)$, and $F$ has mean $F(\pi)/2 > 0$. By uniqueness, $f = 2F/F(\pi)$. On $[0, \pi]$, $F \ge 0$ with maximum $\max(F(s), F(\pi))$. On $[\pi, 2\pi]$, $F = F(\pi) - F(\cdot - \pi) \ge 0$ iff $F(s) \le F(\pi)$, with an extra zero exactly when $F(s) = F(\pi)$. If $F(s) > F(\pi)$, then $F(\pi + s) = F(\pi) - F(s) < 0 < F(\pi)$, so $F$ vanishes somewhere in $(\pi, \pi + s)$, which contains neither 0 nor $L$. Finally $F(s) = I_{\text{in}}$, and $F(\pi) = I_{\text{out}}$.

*Sign pattern.* Suppose $\Delta(y) = 0$ for some $y \in (0, \pi)$. Then $f_{p,q,y} \ge 0$, and its extra root $s+\pi$ lies in $(\pi, \pi + y)$. Given $0 < x < y < z < \pi$, choose $\gamma \in (z, \pi)$ and apply Theorem 13(3) with $a = 0$, $\Gamma = (0, \gamma)$, $m_a = 2p$, $m_b = 2q$; note $(\pi, \pi + y) \subset \Gamma + \pi$. It gives $\Delta(x) > 0$ and $\Delta(z) < 0$. So $\Delta$ has at most one zero, and if $\Delta(y) = 0$ then $\Delta > 0$ before $y$ and $\Delta < 0$ after $y$. By Theorem 13(1), $\Delta > 0$ for small $L$. $\Delta$ is continuous in $L$, because $A$, $s$ and the integrals are. This gives the sign pattern.

$\phi_k = \min L(p, q)$. If $L < \phi_k$, the configuration $\{0, L\}$ lies in an open arc of length $< \phi_k$, so by Theorem 13(1) $\Delta_{p,q}(L) > 0$ for every split, and $L(p, q) \ge \phi_k$. By (2), some split has $\Delta_{p,q}(\phi_k) = 0$. *Symmetry.* $t \mapsto f_{p,q,L}(L - t)$ is raked with constant term 1, so it equals $f_{q,p,L}$. □

## 4. Upper bounds

**Proposition 15** (balanced split)**.** *Let $k$ be even, $p = q = k/2$, and $c = \cos(L/2)$. Then*

$$\Delta(L) = c^{2k-1}B(\tfrac{1}{2}, k) - J_k(c), \qquad J_k(c) := \int_c^1 (x^2 - c^2)^{k-1}\, dx.$$

*Proof.* Here $a = b = k-1$, and $A$ is symmetric about $L/2$, so $s = L/2$. Substituting $u = L/2 - w$, we have $\sin u \sin(L-u) = \cos^2 w - c^2$, and $I_{\text{in}} = \int_0^{L/2}(\cos^2 w - c^2)^{k-1}\sin w\, dw = J_k(c)$. Substituting $u = \frac{L+\pi}{2} + w$, we have $\sin u \sin(u - L) = c^2 - \sin^2 w$ and $\sin(u - s) = \cos w$. Hence $I_{\text{out}} = \int_{-c}^{c}(c^2 - y^2)^{k-1}dy = c^{2k-1}B(\frac{1}{2}, k)$. □

**Lemma 16.** *$J_m(c) \ge (1-c^2)^m/(2m)$ for $c \in (0, 1)$, $m \ge 1$.*

*Proof.* With $x = 1 - y$: $x^2 - c^2 \geq (1 - c^2) - 2y \geq 0$ for $0 \leq y \leq (1 - c^2)/2 \leq 1 - c$. Integrate $((1 - c^2) - 2y)^{m-1}$ over $[0, (1 - c^2)/2]$. □

**Lemma 17** (Wendel [5])**.** *For $x > 0$: $x/\sqrt{x + 1/2} \leq \Gamma(x + \frac{1}{2})/\Gamma(x) \leq \sqrt{x}$.*

*Proof of Theorem 4(i).* By Theorem 3 it suffices to show $\Delta(\frac{\pi}{2} + \delta) < 0$ for $\delta = \delta_+ := \log(8\pi(k + \frac{1}{2}))/(4(k - 1))$, which satisfies $\delta_+ \leq 0.4$ for $k \geq 4$. By Proposition 15 and Lemma 16 it suffices that $c^{2k-1}B(\frac{1}{2}, k) < (1 - c^2)^k/(2k)$, i.e.

$$2k \log \tan \alpha > \log \big(2kB(\tfrac{1}{2}, k)/c\big), \qquad \alpha = \tfrac{\pi}{4} + \tfrac{\delta}{2},\ c = \cos \alpha.$$

We have $\log \tan \alpha = 2 \operatorname{artanh} \tan \frac{\delta}{2} \geq \delta$ and $c = (\cos \frac{\delta}{2} - \sin \frac{\delta}{2})/\sqrt{2} \geq (1 - \delta)/\sqrt{2}$. By Lemma 17, $B(\frac{1}{2}, k) = \sqrt{\pi}\,\Gamma(k)/\Gamma(k + \frac{1}{2}) \leq \sqrt{\pi(k + \frac{1}{2})}/k$. So the right side is at most $\frac{1}{2} \log(8\pi(k + \frac{1}{2})) - \log(1 - \delta) \leq \frac{1}{2} \log(8\pi(k + \frac{1}{2})) + 2\delta$. This is $\leq 2k\delta$ for $\delta \geq \delta_+$. Since $\log \tan \alpha > \delta$ strictly for $\delta > 0$, the required strict inequality holds at $\delta = \delta_+$. □

**Odd $k$: a three-cluster configuration.** Let $k$ be odd and $\alpha \in (0, \pi/2)$. Consider roots of multiplicity $k - 1$ at $\pm\alpha$ and 2 at 0. All multiplicities are even, they sum to $2k$, and they lie in an arc of length $2\alpha$. By Theorem 13(1), if the corresponding $f$ is negative somewhere, then $\phi_k \leq 2\alpha$.

**Proposition 18.** *Let $c = \cos \alpha$, $w(y) = (y^2 - c^2)^{k-2}$ on $[c, 1]$, and $c_\sigma^2 = \int_c^1 wy^2 / \int_c^1 w$. Then $f \geq 0$ on $\mathbb{S}$ iff*

$$\text{LHS} := \int_{c_\sigma}^1 w(y)(y^2 - c_\sigma^2)\, dy \ \leq\ \text{RHS} := 2 \int_0^c (c^2 - y^2)^{k-2}(c_\sigma^2 - y^2)\, dy.$$

*Proof.* By uniqueness $f$ is even. $f'$ is odd-harmonic of degree $\leq 2k - 1$, with roots of multiplicity $k - 2$ at $\pm\alpha$ and 1 at 0, so it is divisible by $(\sin^2 t - \sin^2 \alpha)^{k-2} \sin t$ in the binary-form sense. The quotient $Q$ is a quadratic form, i.e. $Q = A + B \cos 2t + C \sin 2t$, and it is even, so $C = 0$. Rolle's theorem on $(0, \alpha)$ gives a root $\sigma$, so $Q \propto \sin^2 t - \sin^2 \sigma$. With $y = \cos u$: $f(t) = K \int_{\cos t}^1 g$, where $g(y) = (c^2 - y^2)^{k-2}(c_\sigma^2 - y^2)$. The condition $f(\alpha) = 0$ is $\int_c^1 g = 0$, which gives the formula for $c_\sigma$ (recall $k - 2$ is odd). The sign of $g$ is $+$ on $|y| < c$, $-$ on $c < |y| < c_\sigma$, and $+$ on $|y| > c_\sigma$. Also $f(\pi) = 2K \int_0^c g > 0$ equals twice the mean, so $K > 0$. Tracking $\int_x^1 g$ for $x$ from 1 down to $-1$, its only possible negative value is at $x = -c_\sigma$, where it equals $2 \int_0^c g - \int_{c_\sigma}^1 g$, and $\int_{c_\sigma}^1 g = \text{LHS}$. By evenness this settles $[-\pi, 0]$ as well. □

*Proof of Theorem 4(ii).* First, RHS $\leq c^{2k-3} B(\frac{1}{2}, k - 1)$. For LHS: let $Y = y^2$ with $y \sim w/Z$ on $[c, 1]$, $Z = J_{k-1}(c)$. Then LHS $= Z\, \mathbb{E}(Y - \mathbb{E}Y)_+ = \frac{Z}{2} \mathbb{E}|Y - \mathbb{E}Y|$. For any density with maximum $f_{\max}$, $\mathbb{E}|Y - \mathbb{E}Y| \geq 1/(4f_{\max})$, since $\mathbb{P}(|Y - \mathbb{E}Y| < r) \leq 2r f_{\max}$. Here $f_Y(v) = w(\sqrt{v})/(2\sqrt{v}Z) \leq (1 - c^2)^{k-2}/(2cZ)$. With Lemma 16 this gives LHS $\geq cZ^2/(4(1 - c^2)^{k-2}) \geq c(1 - c^2)^k/(16(k - 1)^2)$. So $f$ is negative somewhere if $2k \log \tan \alpha > \log \big(16(k - 1)^2 B(\frac{1}{2}, k - 1)\big) - 4 \log c$. Put $\alpha = \frac{\pi}{4} + \frac{\delta}{2}$ and use $\log \tan \alpha \geq \delta$, $-4 \log c \leq 2 \log 2 + 8\delta$ for $\delta \leq 1/2$, and $B(\frac{1}{2}, k - 1) \leq \sqrt{\pi(k - \frac{1}{2})}/(k - 1)$. It suffices that $(2k - 8)\delta > \log(64\sqrt{\pi}(k - 1)\sqrt{k - 1/2})$. The resulting $\delta$ is $\leq 1/2$ for $k \geq 13$. Strictness: put $\delta_0 = \log(64\sqrt{\pi}(k - 1)\sqrt{k - 1/2})/(2k - 8)$. At $\delta = \delta_0$ the sufficient inequality still holds strictly, because $\log \tan \alpha > \delta$ for $\delta > 0$. So $f$ is negative somewhere at $\alpha_0 = \frac{\pi}{4} + \frac{\delta_0}{2}$. The coefficients of $f$ solve a linear system (constant term 1, and roots of multiplicity at least $k - 1, 2, k - 1$ at $-\alpha, 0, \alpha$) whose matrix is continuous in $\alpha$. It is invertible: a kernel vector would be a nonzero raked polynomial with zero constant term and at least $2k$ roots in the open semicircle $(-\frac{\pi}{2}, \frac{\pi}{2})$, which Theorem 12(a) excludes. So $f$ depends continuously on $\alpha$, and negativity is an open condition in $\alpha$. Hence $f$ is negative somewhere at some $\alpha' < \alpha_0$, and $\phi_k \leq 2\alpha' < \frac{\pi}{2} + \delta_0$. □

**Odd $k$: the sharp order of magnitude.** For odd $k$ the three-cluster bound loses a factor 3 in front of $\log k$. The near-balanced split $p=\frac{k-1}{2}$, $q=\frac{k+1}{2}$ recovers the factor 1. The tools are those of §5 below, used in the opposite direction.

**Proposition 19** (general-split upper criterion)**.** *Let $2\le p<q$ with $b-a=2$, and put $N=a+b$, $L=\frac{\pi}{2}+\delta$, $t=\tan\delta$, $w=N^{-1/3}$. Assume $\frac{L}{2}-\frac{2}{a}-w>0$, $\frac{L}{2}+\frac{2}{a}+w\le\frac{\pi}{2}$, and $\kappa:=(\sqrt{a}+\sqrt{b})^2/(1+\sin\delta)>4$. Put*

$$K_w=\frac{a}{\sin^2(\frac{L}{2}-\frac{2}{a}-w)}+\frac{b}{\sin^2(\frac{L}{2}-w)},\qquad E_{\mathrm{low}}=\tfrac14\sqrt{2\pi/K_w}\,\big(1-\mathrm{erfc}(w\sqrt{K_w/2})\big)-\tfrac16(152/\kappa^2)^{3/4}.$$

*Let $D_1=2\Gamma(p+\frac12)\Gamma(q+\frac12)/(\Gamma(p)\Gamma(q))$ as in Lemma 24(c). If $E_{\mathrm{low}}>0$ and $(\delta+t)D_1+b\log\cos\delta-\frac{bt^2p}{2(q-1)}>\log 2-\log E_{\mathrm{low}}$, then $\Delta_{p,q}(L)<0$, and hence $\phi_k\le L(p,q)<L$.*

*Proof.* We use the notation $M_{x,y}(\ell)=\int_0^\ell \sin^x u\,\sin^y(\ell-u)\,du$ and Lemmas 22 and 24 of §5 below; their proofs do not use this subsection. *Outer integral.* $I_{\mathrm{out}}\le M_{a,b}(\pi-L)$ since $\sin\le 1$, and by log-concavity (Lemma 24(a),(c)) $\log M_{a,b}(\frac{\pi}{2}-\delta)\le\log M_{a,b}(\frac{\pi}{2})-\delta D_1$.

*Inner mass.* Dropping $u\in(\frac{\pi}{2},L)$, we get $M_{a,b}(L)\ge\cos^b\delta\,M_{a,b}(\frac{\pi}{2})\,\mathbb{E}_0(1+tT)^b$, where $u\sim\sin^a u\cos^b u$ on $(0,\frac{\pi}{2})$ and $T=\tan u$. By Jensen and $\log(1+x)\ge x-x^2/2$, $\mathbb{E}_0(1+tT)^b\ge\exp(tb\mathbb{E}_0T-\frac{bt^2}{2}\mathbb{E}_0T^2)$, with $b\mathbb{E}_0T=bB(p+\frac12,q-\frac12)/B(p,q)=D_1$ (as in the proof of Lemma 24(c)) and $\mathbb{E}_0T^2=B(p+1,q-1)/B(p,q)=p/(q-1)$.

*Mean absolute deviation.* $I_{\mathrm{in}}=\frac12 M_{a,b}(L)\,\mathbb{E}|\sin(u-s)|$ for $u\sim A/M$ on $(0,L)$, $A=e^{-V}$. Let $u_1$ be the mode, so $V'(u_1)=0$. Since $V'(\frac{L}{2})=(b-a)\cot\frac{L}{2}>0$ and $V'(\frac{L}{2}-\eta)\le-2a\eta+(b-a)$ (because $\cot(x-\eta)-\cot(x+\eta)\ge2\eta$ and $\cot(\frac{L}{2}+\eta)\le1$), we have $u_1\in(\frac{L}{2}-\frac{2}{a},\frac{L}{2})$. On $|u-u_1|\le w$ we have $V''\le K_w$ by the two assumptions on the window. Hence $A(u)\ge A(u_1)e^{-K_w(u-u_1)^2/2}$ there, and the maximal density satisfies $A(u_1)/M\le 1/\int_{-w}^{w}e^{-K_wx^2/2}dx$. Since $\mathbb{P}(|u-s|<r)\le 2rA(u_1)/M$, $\mathbb{E}|u-s|\ge M/(4A(u_1))$. Integrating $(u-u_1)^{2j-1}V'(u)$ by parts, with $V''\ge\kappa$ (proof of Lemma 22), gives $\mathbb{E}(u-u_1)^2\le1/\kappa$ and $\mathbb{E}(u-u_1)^4\le3/\kappa^2$. Hence $\mathbb{E}\cos(u-u_1)\ge1-\frac{1}{2\kappa}\ge\frac12$ and $|\mathbb{E}\sin(u-u_1)|\le\mathbb{E}|u-u_1|\le\kappa^{-1/2}<\frac12$, since $\kappa>4$. The angle $s$ maximizes $\sigma\mapsto\mathbb{E}\cos(u-\sigma)$, so $\mathbb{E}\cos(u-s)\ge\mathbb{E}\cos(u-u_1)$. If $|s-u_1|\ge\frac{\pi}{2}$, then $\mathbb{E}\cos(u-s)=\cos(s-u_1)\mathbb{E}\cos(u-u_1)+\sin(s-u_1)\mathbb{E}\sin(u-u_1)\le|\mathbb{E}\sin(u-u_1)|<\frac12\le\mathbb{E}\cos(u-u_1)$, a contradiction. So $|s-u_1|<\frac{\pi}{2}$. From $\mathbb{E}\sin(u-s)=0$ we get $\tan(s-u_1)=\mathbb{E}\sin(u-u_1)/\mathbb{E}\cos(u-u_1)$, hence $|s-u_1|\le|\tan(s-u_1)|\le\kappa^{-1/2}/(1-\frac{1}{2\kappa})\le2\kappa^{-1/2}$. Thus $\mathbb{E}(u-s)^4\le8(3+16)/\kappa^2$. Finally $\mathbb{E}|\sin(u-s)|\ge\mathbb{E}|u-s|-\frac16\mathbb{E}|u-s|^3\ge E_{\mathrm{low}}$. Combining, $\log I_{\mathrm{in}}-\log I_{\mathrm{out}}\ge(\delta+t)D_1+b\log\cos\delta-\frac{bt^2p}{2(q-1)}-\log2+\log E_{\mathrm{low}}>0$. □

*Proof of Theorem 6.* Put $C=4.1$ and $\delta_k=(\log k+C)/(4k)$.

$2\le k\le300$. The enclosures of Theorem 8 give $\phi_k<\frac{\pi}{2}+\delta_k$, compared in ball arithmetic. The smallest margin is $7.2\cdot10^{-4}$, at $k=300$ (`numerics/bln_odd_upper_arb.py`, output `numerics/odd_upper_arb.txt`).

*Even $k\ge302$.* By Theorem 4(i), $4k(\phi_k-\frac{\pi}{2})-\log k<g(k):=\log(8\pi)+\log(1+\frac{1}{2k})+\frac{\log(8\pi(k+1/2))}{k-1}$. Both $k$-dependent terms decrease in $k$; for the last one,

$$\frac{d}{dx}\,\frac{\log(8\pi(x+\frac12))}{x-1}=\frac{(x-1)/(x+\frac12)-\log(8\pi(x+\frac12))}{(x-1)^2}<0.$$

Hence $g(k)\le g(302)<3.26<C$.

*Odd $301\le k\le200001$.* Apply Proposition 19 with $p=\frac{k-1}{2}$, $q=\frac{k+1}{2}$ (so $a=k-2$, $b=k$) and $\delta=\delta_k$. For each of these 99851 values of $k$, its four hypotheses and its final inequality

were certified in Arb ball arithmetic (precision 96 bits). The smallest certified margin in the final inequality is 0.0257, at $k = 301$ (same script).

*Odd $k \geq 200003$.* We verify the hypotheses and the inequality of Proposition 19 by hand. Let $N = 2k-2$ and $w = N^{-1/3}$, and put $\varepsilon_1 = \frac{2}{a} + w$. Then $\delta := \delta_k \leq 2.04 \cdot 10^{-5}$ and $\varepsilon_1 \leq 0.0135821$, and both decrease in $k$. So $\frac{L}{2} - \frac{2}{a} - w = \frac{\pi}{4} + \frac{\delta}{2} - \varepsilon_1 > 0$ and $\frac{L}{2} + \frac{2}{a} + w < \frac{\pi}{2}$; and $\kappa \geq 4a/(1+\delta) > 4$, since $b > a$. Both arguments of sin in $K_w$ lie in $[\frac{\pi}{4} - \varepsilon_1, \frac{\pi}{2}]$. Moreover $\sin(\frac{\pi}{4} - \varepsilon) = (\cos\varepsilon - \sin\varepsilon)/\sqrt{2} \geq s(\varepsilon)/\sqrt{2}$, where $s(\varepsilon) = 1 - \varepsilon - \varepsilon^2/2$. Hence $K_w \leq 2N/s(\varepsilon_1)^2$, and $\frac{1}{4}\sqrt{2\pi/K_w} \geq \frac{1}{4}\sqrt{\pi/N}\, s(\varepsilon_1)$. Since $\sin^2 \leq 1$, $K_w \geq a + b = N$, so $\operatorname{erfc}(w\sqrt{K_w/2}) \leq e_1 := \operatorname{erfc}(N^{1/6}/\sqrt{2}) < 10^{-16}$. From $\kappa \geq 4a/(1+\delta)$ we get $\frac{1}{6}(152/\kappa^2)^{3/4} \leq 0.902\, a^{-3/2} \leq \frac{1}{4}\sqrt{\pi/N} \cdot \frac{2.9}{a}$, using $N = 2a+2$. Therefore

$$E_{\text{low}} \geq \tfrac{1}{4}\sqrt{\pi/N}\,(1-\eta), \qquad \eta := \varepsilon_1 + \tfrac{1}{2}\varepsilon_1^2 + e_1 + \tfrac{2.9}{a} \leq 0.01369.$$

In particular $E_{\text{low}} > 0$, and since $N < 2k$, $\log 2 - \log E_{\text{low}} < \frac{1}{2}\log k + \frac{1}{2}\log\frac{128}{\pi} - \log(1-\eta)$. On the other side, Lemma 17 gives $D_1 \geq 2pq/\sqrt{(p+\frac{1}{2})(q+\frac{1}{2})} = (k^2-1)/\sqrt{k(k+2)} \geq k-1$. Since $\delta \leq t \leq 2\delta$, $\log\cos\delta \geq -\delta^2$ and $q - 1 = p$, the left side of the inequality in Proposition 19 is at least $2\delta(k-1) - k\delta^2 - \frac{k}{2}t^2 \geq \frac{1}{2}(\log k + C) - 2\delta - 3k\delta^2$. The quantity $3k\delta^2 = 3(\log k + C)^2/(16k)$ decreases in $k$ and is $\leq 2.5 \cdot 10^{-4}$. So the margin in the inequality of Proposition 19 is at least

$$\tfrac{C}{2} - \tfrac{1}{2}\log\tfrac{128}{\pi} + \log(1-\eta) - 2\delta - 3k\delta^2 \;\geq\; 2.05 - 1.85366 - 0.01379 - 0.00005 - 0.00025 \;>\; 0.18.$$

In all cases Proposition 19 or the preceding cases give $\phi_k < \frac{\pi}{2} + \delta_k$. □

**Remark 20.** Letting $C \downarrow \log(128/\pi)$ in the last step shows that Proposition 19 alone gives $\limsup_{k \text{ odd}} \big(4k(\phi_k - \frac{\pi}{2}) - \log k\big) \leq \log(128/\pi) = 3.707$. Theorem 7 gives the sharp value $\log(8\pi) = 3.224$ for both parities. The explicit lower bounds on the margin used in the last step are compared with the exact criterion at $k = 200003$, $10^6+1$, $10^8+1$ and $10^{12}+1$ in the same script.

## 5. Lower bound: proof of Theorem 5

Throughout, $p \leq q$ (by symmetry), $a = 2p-1$, $b = 2q-1$, $L = \frac{\pi}{2} + \delta$ with $0 < \delta \leq 0.2$, $t = \tan\delta$, and $M_{x,y}(\ell) = \int_0^\ell \sin^x u\, \sin^y(\ell - u)\, du$.

**Lemma 21** (variance)**.** *Let $\mu = e^{-V}du$ be a probability measure on an interval, $V \in C^2$, $V'' \geq \kappa > 0$, and $(u-m)e^{-V} \to 0$ at the endpoints, where $m = \int u\, d\mu$. Then $\operatorname{Var}_\mu(u) \leq 1/\kappa$. If moreover the interval has length $< \pi$ and $s = \arg\int e^{iu} d\mu$, then $\int \sin^2(u-s)\, d\mu \leq \operatorname{Var}_\mu(u)$.*

*Proof.* Integrating by parts, $\int (u-m)V'(u)\, d\mu = 1$. Also $\int (u-m)V'(m)\, d\mu = 0$, and $(u-m)(V'(u) - V'(m)) \geq \kappa(u-m)^2$. Hence $\kappa \operatorname{Var} \leq 1$. Next, $\sigma \mapsto \int \cos(u-\sigma) d\mu = |\int e^{iu} d\mu| \cos(\sigma - s)$ is maximal at $\sigma = s$. Hence $\int \sin^2(u-s) \leq 2\int(1 - \cos(u-s)) \leq 2\int(1-\cos(u-m)) \leq \int (u-m)^2$. □

**Lemma 22.** $I_{\text{in}} \leq \frac{1}{2} M_{a,b}(L)\, \kappa_\delta^{-1/2}$ *with* $\kappa_\delta = (\sqrt{a} + \sqrt{b})^2/(1+\sin\delta)$.

*Proof.* Apply Lemma 21 to $A/M_{a,b}(L) = e^{-V}$, with $V'' = a/\sin^2 u + b/\sin^2(L-u)$. By Cauchy–Schwarz, $V''$ is at least $(\sqrt{a}+\sqrt{b})^2/(\sin^2 u + \sin^2(L-u))$, and $\sin^2 u + \sin^2(L-u) = 1 - \cos L \cos(2u - L) \leq 1 + \sin\delta$. Then $I_{\text{in}} = \frac{1}{2} M \int |\sin(u-s)| \leq \frac{1}{2} M (\int \sin^2(u-s))^{1/2}$. □

**Lemma 23.** $I_{\text{out}} \geq M_{a+1,b+1}(\pi - L)$.

*Proof.* For $u \in [L, \pi]$ we have $u - s \in [u-L, u] \subset [0, \pi]$, so by concavity of sin, $\sin(u-s) \geq \min(\sin u, \sin(u-L)) \geq \sin u \sin(u-L)$. Substitute $v = u - L$ and use $\sin(L+v) = \sin(\pi - L - v)$. □

**Lemma 24.** *(a) $\ell \mapsto M_{x,y}(\ell)$ is log-concave on $(0,\pi)$ for $x,y \geq 0$. (b) $M_{a,b}(\pi/2) = \frac{1}{2}B(p,q)$, $M_{a+1,b+1}(\pi/2) = \frac{1}{2}B(p+\frac{1}{2}, q+\frac{1}{2})$. (c) $D_1 := (\log M_{a,b})'(\pi/2) = 2\frac{\Gamma(p+1/2)\Gamma(q+1/2)}{\Gamma(p)\Gamma(q)} \leq 2\sqrt{pq}$. (d) Let $u \sim \sin^{a+1} u \cos^{b+1} u\, du$ on $(0,\pi/2)$ and $T = \tan u$. Then $(b+1)\mathbb{E}T = D_2 := 2\frac{\Gamma(p+1)\Gamma(q+1)}{\Gamma(p+1/2)\Gamma(q+1/2)}$, and $\sin^2 u \sim \mathrm{Beta}(p+\frac{1}{2}, q+\frac{1}{2})$.*

*Proof.* (a) $M_{x,y} = h_x * h_y$ with $h_x = \sin^x \cdot 1_{(0,\pi)}$ log-concave, and a convolution of log-concave functions is log-concave (Prékopa's theorem [6]). (b),(d) are Beta integrals: $\int_0^{\pi/2} \sin^x u \cos^y u\, du = \frac{1}{2}B(\frac{x+1}{2}, \frac{y+1}{2})$. (c) $M'_{a,b}(\ell) = b\int \sin^a u \sin^{b-1}(\ell-u)\cos(\ell-u)$, which at $\ell = \pi/2$ gives $bB(p+\frac{1}{2}, q-\frac{1}{2})/B(p,q)$; simplify, and use Lemma 17. ☐

**Proposition 25** (explicit safety criterion)**.** *Let $\tau = (4/5)^{p-1/2}(1/5)^{q+1/2}/B(p+\frac{1}{2}, q+\frac{1}{2})$ and suppose $\tau < 1$, $t < 1/2$. Define*

$$\Phi(p,q,\delta) = \log\tfrac{B(p,q)}{B(p+\frac{1}{2},q+\frac{1}{2})} + \delta D_1 + \frac{tD_2}{(1-2t)(1-\tau)} - (b+1)\log\cos\delta - \log(1-\tau) - \log 2 - \tfrac{1}{2}\log\kappa_\delta.$$

*If $\Phi(p,q,\delta) < 0$, then $\Delta_{p,q}(\frac{\pi}{2}+\delta) > 0$, and hence $L(p,q) > \frac{\pi}{2}+\delta$.*

*Proof.* By Lemmas 22 and 23 it suffices that $\log M_{a,b}(\frac{\pi}{2}+\delta) - \log M_{a+1,b+1}(\frac{\pi}{2}-\delta) < \log 2 + \frac{1}{2}\log\kappa_\delta$. By Lemma 24(a), $\log M_{a,b}(\frac{\pi}{2}+\delta) \leq \log M_{a,b}(\frac{\pi}{2}) + \delta D_1$. Next, $M_{a+1,b+1}(\frac{\pi}{2}-\delta) = \int_0^{\pi/2-\delta} \sin^{a+1} u \cos^{b+1}(u+\delta)du$, and $\cos(u+\delta) = \cos\delta\cos u(1-tT)$. So $M_{a+1,b+1}(\frac{\pi}{2}-\delta) = \cos^{b+1}\delta\, M_{a+1,b+1}(\frac{\pi}{2})\,\mathbb{E}(1-tT)_+^{b+1}$. On $\{T \leq 2\}$, $\log(1-tT) \geq -tT/(1-2t)$. By Jensen on the conditional law, $\mathbb{E}(1-tT)_+^{b+1} \geq P_0 \exp(-(b+1)t\mathbb{E}T/((1-2t)P_0))$ with $P_0 = \mathbb{P}(T \leq 2)$. Moreover $1 - P_0 = \mathbb{P}(\sin^2 u > 4/5) \leq \tau$: the $\mathrm{Beta}(p+\frac{1}{2}, q+\frac{1}{2})$ density is decreasing on $[4/5, 1]$ because $p \leq q$. Combine with Lemma 24(b),(d). ☐

*Proof of Theorem 5(i).* Fix $c < 2\log 2$ and put $\delta = (\log k + c)/(4k)$. We show $\max_{p\leq q}\Phi < 0$ for large $k$. Lemma 17 gives $\log\frac{B(p,q)}{B(p+1/2,q+1/2)} \leq \log k - \frac{1}{2}\log(pq) + \frac{1}{2}\log\big((1+\frac{1}{2p})(1+\frac{1}{2q})\big)$, $D_2 \leq 2\sqrt{(p+\frac{1}{2})(q+\frac{1}{2})}$, and $D_1 \leq 2\sqrt{pq}$. Also $\tau \leq C\sqrt{k}\,(0.8)^k$ uniformly in $p \leq q$, using $1/B(\alpha,\beta) \leq 2^{\alpha+\beta-1}\sqrt{(\alpha+\beta)/(2\pi)}$ (log-convexity of $\Gamma$ and duplication). Further $-\log\cos\delta \leq \delta^2$, and $\frac{t}{\delta(1-2t)(1-\tau)} = 1 + O(\delta+\tau)$. Finally $\delta\sqrt{pq}\,(\frac{1}{p}) \leq \delta\sqrt{k} \to 0$, $k\delta^2 \to 0$, and $\delta k\,(\delta+\tau) \to 0$. So, uniformly in $p$,

$$\Phi \leq F + o(1), \qquad F = -\log y + (\log k + c)y - \log 2 - \log(\sqrt{a}+\sqrt{b}) + \tfrac{1}{2}\log(1+\tfrac{1}{2p}), \qquad y = \tfrac{\sqrt{pq}}{k} \in (0, \tfrac{1}{2}].$$

Since $(\sqrt{a}+\sqrt{b})^2 = 2k - 2 + 2\sqrt{(2p-1)(2q-1)} \geq 2k(1+2y) - 2 - 2/y$ and $1/y \leq k/\sqrt{k-1}$, we have $\log(\sqrt{a}+\sqrt{b}) \geq \frac{1}{2}\log(2k) + \frac{1}{2}\log(1+2y) - o(1)$. Thus $F \leq h(y) + \frac{1}{2}\log(1+\frac{1}{2p}) + o(1)$, where $h(y) = -\log y + (\log k + c)y - \frac{1}{2}\log k - \frac{3}{2}\log 2 - \frac{1}{2}\log(1+2y)$ is convex. We have $h(\frac{1}{2}) = \frac{c}{2} - \log 2 < 0$, $h(y_{\min}) = -\frac{3}{2}\log 2 + o(1)$ with $y_{\min} = \sqrt{k-1}/k$, and $h(k^{-1/4}) = -\frac{1}{4}\log k - \frac{3}{2}\log 2 + o(1)$. On $[k^{-1/4}, \frac{1}{2}]$, $p \geq ky^2 \geq \sqrt{k}$, so the extra term is $o(1)$, and by convexity $h \leq \max(h(k^{-1/4}), h(\frac{1}{2})) < 0$. On $[y_{\min}, k^{-1/4}]$, the extra term is at most $\frac{1}{2}\log\frac{3}{2} < 0.203$ and $h \leq -\frac{3}{2}\log 2 + o(1) < -1.03$. Hence $\Phi < 0$ for all splits when $k$ is large. Proposition 25 and Theorem 3 give $\phi_k > \frac{\pi}{2} + \delta$ for large $k$. ☐

*Proof of Theorem 5(ii).* Take $c = 0$. For $k > 2 \cdot 10^4$ the error terms in the previous proof are explicitly small. Here $\delta \leq 1.3 \cdot 10^{-4}$, $(1+3\delta)\delta\sqrt{k}/2 \leq 0.009$ (the $1/p$ part of the $D_2$ correction, including the factor $\frac{t}{(1-2t)(1-\tau)} \leq (1+3\delta)\delta$), $3k\delta^2 \leq 10^{-3}$ (the $O(\delta)$ part), $2k\delta^2 = (\log k)^2/(8k) \leq 7 \cdot 10^{-4}$ (the $\log\cos\delta$ term), the $\log(\sqrt{a}+\sqrt{b})$ correction is $\leq (1 + k/\sqrt{k-1})/k < 0.008$, and $\tau < 10^{-900}$. The remaining terms are also small: $\frac{1}{2}\log(1+\sin\delta) \leq \delta/2$; $\frac{1}{2}\log(1+\frac{1}{2q}) \leq \frac{1}{2k}$ (as $q \geq k/2$); the $1/q$ and $1/(pq)$ parts of the $D_2$ correction, at most $\frac{t}{(1-2t)(1-\tau)} \cdot 2\sqrt{pq}\,(\frac{1}{4q} + \frac{1}{16pq}) \leq \delta$;

and $-\log(1-\tau) \le 2\tau$. Together they contribute $< 3 \cdot 10^{-4}$. All these bounds decrease in $k$, and their total is $< 0.02$. On the other side, $h(\frac{1}{2}) = -\log 2$, $h(k^{-1/4}) \le -2.7$, and on the small-$p$ branch $h \le h(y_{\min}) \le -0.976$, so even after adding $\frac{1}{2}\log\frac{3}{2}$ it stays $\le -0.77$. On the large-$p$ branch, $p \ge \sqrt{k}$ and the extra term $\frac{1}{2}\log(1+\frac{1}{2p})$ is $\le \frac{1}{4\sqrt{k}} < 0.002$. So in every case the error total is $< 0.025$, against a margin of at least 0.69. Hence $\Phi < 0$.

For $301 \le k \le 2 \cdot 10^4$ we certified $\Phi(p, k-p, \frac{\log k}{4k}) < 0$, together with the hypotheses $\tau < 1$ and $t < \frac{1}{2}$ of Proposition 25, in Arb ball arithmetic (precision 96 bits) for each of the 99977500 pairs $(k,p)$ with $p \le k/2$. All Gamma arguments are multiples of $\frac{1}{2}$, so $\log\Gamma(n/2)$ is tabulated once. The largest upper endpoint of the certified enclosures of $\Phi$ is $-0.6127$, at $k = 301$, $p = 1$ (`numerics/bln_lower_arb.py`, output `numerics/lower_arb.txt`; about 80 seconds on 8 cores). For $2 \le k \le 300$, the enclosures of Theorem 8 give $\phi_k - \frac{\pi}{2} - \frac{\log k}{4k} > 0$; the smallest margin is 0.0026, at $k = 300$ (same script). □

## 6. Sharp asymptotics: proof of Theorem 7

**Lemma 26** (Laplace bounds for strongly log-concave laws)**.** *Let $\mu = e^{-V}du/Z$ be a probability measure on an open interval $J$, $V \in C^2(J)$, $V'' \ge \kappa > 0$, and $e^{-V} \to 0$ at the endpoints. Let $V'(u_1) = 0$, $K = V''(u_1)$, $y = u - u_1$. Suppose $|V'' - K| \le \eta K$ on $[u_1 - w, u_1 + w] \subset J$ with $\eta \le \frac{1}{2}$, and put $\lambda = (1-\eta)Kw^2/2 \ge 1$. Then, with an absolute constant $C$:*

$$\left|\mathbb{E}|y| - \sqrt{2/(\pi K)}\right| \le C(\eta + e^{-\lambda})K^{-1/2}, \qquad |\mathbb{E}y| \le C(\eta+e^{-\lambda})K^{-1/2}, \qquad \mathbb{E}y^2 \le \kappa^{-1}, \quad \mathbb{E}y^4 \le 3\kappa^{-2}.$$

*Proof.* Let $\varphi(y) = V(u_1+y) - V(u_1)$ and $K_\pm = (1 \pm \eta)K$. For $|y| \le w$, Taylor's formula gives $K_- y^2/2 \le \varphi \le K_+ y^2/2$. For $|y| > w$, convexity and $\varphi'(\pm w) \gtrless \pm K_- w$ give $\varphi(y) \ge K_- w(|y| - w/2)$. Hence the parts of $\int e^{-\varphi}$ and $\int |y| e^{-\varphi}$ over $|y| > w$ are at most $\frac{2}{K_- w}e^{-\lambda}$ and $\frac{2e^{-\lambda}}{K_- w}(w + \frac{1}{K_- w})$. Over $|y| \le w$ they lie between the corresponding Gaussian integrals with $K_+$ and $K_-$, namely $\sqrt{2\pi/K_\pm}\,(1 - O(e^{-\lambda}))$ and $\frac{2}{K_\pm}(1 - O(e^{-\lambda}))$. Taking ratios gives the first bound. For the second, compare $\int_0^w y e^{-\varphi}$ and $\int_{-w}^0 |y| e^{-\varphi}$; both lie in $[\frac{1}{K_+}(1-e^{-\lambda}), \frac{1}{K_-}]$. The last two bounds come from integrating $y^{2j-1}V'(u_1+y) \ge \kappa y^{2j}$ by parts ($j = 1, 2$). □

Fix $\epsilon = \frac{1}{5}$ and call a split *central* if $\epsilon k \le p \le q$ and *extreme* if $p < \epsilon k$ (with $p \le q$). Throughout, $o(1)$ is uniform over central splits as $k \to \infty$, and $0 \le \delta \le 2\log k/k$, $L = \frac{\pi}{2} + \delta$, $L' = \pi - L$. Write $u_1(\ell)$ for the mode of $A_\ell = \sin^a u \sin^b(\ell - u)$ on $(0, \ell)$, and $r_{ab} = 2\sqrt{ab}/(a+b)$.

**Lemma 27** (central splits)**.** *(a) $\tan^2 u_1(\frac{\pi}{2}) = a/b$, $|u_1'(\ell)| \le 1$, and $K(\ell) := V_\ell''(u_1(\ell)) = 2N(1 + O(|\ell - \frac{\pi}{2}|))$. Moreover $|V_\ell'''| = O(N)$ within distance 0.1 of $u_1(\ell)$. (b) $I_{\text{in}} = \frac{1}{2}M_{a,b}(L)(\pi N)^{-1/2}(1 + o(1))$. (c) $I_{\text{out}} = M_{a,b}(L')\, r_{ab}(1 + o(1))$. (d) $\Psi := \log M_{a,b}(L) - \log M_{a,b}(L') = 2\delta D_1 + O(N\delta^2) + o(1)$, and $D_1 = 2\sqrt{pq}(1 + O(1/k))$.*

*Proof.* (a) $u_1$ solves $F(u,\ell) := a\cot u - b\cot(\ell - u) = 0$, and $u_1' = -\partial_\ell F/\partial_u F = \frac{b\csc^2(\ell-u)}{a\csc^2 u + b\csc^2(\ell-u)} \in (0,1)$. At $\ell = \frac{\pi}{2}$, $\sin^2 u_1 = a/N$, so $K = a \cdot \frac{N}{a} + b \cdot \frac{N}{b} = 2N$. For central splits $\tan^2 u_1(\frac{\pi}{2}) \in [\frac{1}{8}, 1]$, so $u_1$ and $\ell - u_1$ stay at distance $\ge 0.25$ from 0 for $|\ell - \frac{\pi}{2}| \le 0.05$. On that range $\csc^2$ and $\csc^2 \cdot \cot$ are bounded, which gives the remaining claims. (b) Apply Lemma 26 to $A_L/M$, with $\kappa \ge N/2$ (Lemma 22), $w = N^{-1/3}$, $\eta = O(N^{-1/3})$ and $\lambda \asymp N^{1/3}$. We get $\mathbb{E}|y| = (\pi N)^{-1/2}(1 + o(1))$ and $\mathbb{E}y = o(N^{-1/2})$. From $\mathbb{E}\sin(u-s) = 0$, $\tan(s - u_1) = \mathbb{E}\sin y/\mathbb{E}\cos y$, and $|\mathbb{E}\sin y - \mathbb{E}y| \le \mathbb{E}|y|^3 = O(N^{-3/2})$. To pass to $s - u_1$ itself, note that $\mathbb{E}\cos y \ge 1 - \frac{1}{2}\mathbb{E}y^2 \ge 1 - \frac{1}{N}$. Also $s$ maximizes $\sigma \mapsto \mathbb{E}\cos(u-\sigma)$, so $\mathbb{E}\cos(u-s) \ge \mathbb{E}\cos y$. If $|s - u_1| \ge \frac{\pi}{2}$, then $\mathbb{E}\cos(u-s) \le |\mathbb{E}\sin y| \le \mathbb{E}|y| < \mathbb{E}\cos y$, a contradiction. Hence $|s - u_1| < \frac{\pi}{2}$, and $s - u_1 = \arctan(\mathbb{E}\sin y/\mathbb{E}\cos y) = o(N^{-1/2})$. Hence $\mathbb{E}|\sin(u-s)| = \mathbb{E}|y| + O(|s-u_1| + \mathbb{E}|y|^3 + |s-u_1|^3) = (\pi N)^{-1/2}(1+o(1))$, and $I_{\text{in}} = \frac{1}{2}M\,\mathbb{E}|\sin(u-s)|$.

(c) The substitution $v=\pi-u$ maps the outer density to $A_{L'}$ and $\sin(u-s)$ to $\sin(v+s)$. By (a) and Lemma 26, $\mathbb{E}|v-u_1(L')| = O(N^{-1/2})$. Also $v_1+s = u_1(L')+u_1(L)+o(1) = 2u_1(\frac{\pi}{2})+O(\delta)+o(1)$, and $\sin 2u_1(\frac{\pi}{2}) = \frac{2\sqrt{a/b}}{1+a/b} = r_{ab}$, and $r_{ab}\geq 0.79$ for central splits once $k$ is large (the limit value at $p=\epsilon k$ is 0.8). Hence $\mathbb{E}\sin(v+s) = r_{ab}(1+o(1))$. (d) Upper bound: $\log M(\frac{\pi}{2}+\delta)\leq \log M(\frac{\pi}{2})+\delta D_1$ by concavity. Also $M(\frac{\pi}{2}-\delta) = \cos^b\delta\, M(\frac{\pi}{2})\,\mathbb{E}_0(1-tT)_+^b$, and on $\{T\leq 1/(2t)\}$, $(1-tT)^b \geq e^{-btT-bt^2T^2}$. By Jensen, with $b\mathbb{E}_0T = D_1$, $\mathbb{E}_0T^2 = \frac{p}{q-1}\leq 2$, and $\mathbb{P}(T>\frac{1}{2t})\leq \mathbb{E}_0T^b(2t)^b$ super-polynomially small, we get $\log M(\frac{\pi}{2}-\delta)\geq \log M(\frac{\pi}{2})-tD_1-O(N\delta^2)$. Lower bound: by concavity $\log M(\frac{\pi}{2}-\delta)\leq \log M(\frac{\pi}{2})-\delta D_1$; and as in Proposition 19, $\log M(\frac{\pi}{2}+\delta)\geq \log M(\frac{\pi}{2})+tD_1-O(N\delta^2)$. Since $t-\delta = O(\delta^3)$, (d) follows; the bound on $D_1$ is Lemma 17. □

*Proof of Theorem 7.* By Theorem 3, a split is safe at $L$ iff $\log I_{\text{out}} > \log I_{\text{in}}$. For central splits, by Lemma 27, this is iff $\Psi < \log(2r_{ab}\sqrt{\pi N})+o(1)$. Put $\delta = (\log k+c)/(4k)$ and $r = 2\sqrt{pq}/k\in[0.8,1]$. Then $r_{ab} = r(1+O(1/k))$, $N=2k-2$, and $N\delta^2 = o(1)$. So the condition reads

$$(*)\qquad \tfrac{r}{2}(\log k + c) < \tfrac{1}{2}\log k + \tfrac{1}{2}\log(8\pi) + \log r + o(1).$$

*Upper bound.* The balanced split ($p=\lfloor k/2\rfloor$) has $r = 1-O(k^{-2})$. By $(*)$ it is unsafe for $c>\log 8\pi$ and $k$ large. Hence $\phi_k \leq L(\lfloor\frac{k}{2}\rfloor,\lceil\frac{k}{2}\rceil)\leq \frac{\pi}{2}+\frac{\log k+\log 8\pi+o(1)}{4k}$. *Lower bound, central splits.* Let $c=\log 8\pi-\epsilon'$. Using $\log r\geq -(1-r)/r\geq -\frac{5}{4}(1-r)$, the right side of $(*)$ minus the left side is at least $(1-r)\big(\frac{1}{2}\log k+\frac{1}{2}\log 8\pi-\frac{5}{4}\big)+\frac{r}{2}\epsilon'-o(1)\geq 0.4\epsilon'-o(1)>0$. So every central split is safe for large $k$. *Lower bound, extreme splits.* The proof of Theorem 5(i) applies verbatim with $c=\log 8\pi$ in the range $y=\sqrt{pq}/k\leq\sqrt{p/k}<\sqrt{\epsilon}$. There $h(\sqrt{\epsilon}) = (\sqrt{\epsilon}-\frac{1}{2})\log k+O(1)\to-\infty$ and $h(y_{\min})\to-\frac{3}{2}\log 2$, so by convexity $\Phi<0$ for large $k$, uniformly over extreme splits. Hence $\phi_k = \min_p L(p,k-p)\geq \frac{\pi}{2}+\frac{\log k+\log 8\pi-\epsilon'}{4k}$ for large $k$, for every $\epsilon'>0$. □

## 7. All sufficiently large $k$: proof of Theorem 9

Write $L_{\text{bal}} = L(\lfloor\frac{k}{2}\rfloor,\lceil\frac{k}{2}\rceil)$. By symmetry we only consider $p\leq q$.

**Lemma 28** (far-from-balanced splits)**.** *There is $K_A$ such that for $k\geq K_A$ every split with $p\leq q$ and $p\leq 0.41k$ satisfies $L(p,k-p)>L_{\text{bal}}$.*

*Proof.* By the upper-bound part of the proof of Theorem 7, $L_{\text{bal}}<\frac{\pi}{2}+\frac{\log k+\log 8\pi+1/40}{4k}$ for large $k$. Let $c_+ = \log 8\pi+\frac{1}{20}$ and $\delta_+ = (\log k+c_+)/(4k)$. For a central split ($p\geq k/5$) with $p\leq 0.41k$ we have $r = 2\sqrt{pq}/k\in[0.8,0.984]$. At $c=c_+$ the right side of $(*)$ minus the left side equals $\frac{1-r}{2}(\log k+\log 8\pi)-\frac{r}{40}+\log r\geq 0.008\log k-\frac{1}{4}$. This tends to $\infty$, so the split is safe at $\delta_+$ for large $k$. For extreme splits ($p<k/5$) we repeat the proof of Theorem 5(i) with $c=c_+$, in the range $y=\sqrt{pq}/k<0.4$. The restriction $c<2\log 2$ in that proof is used only at $y=\frac{1}{2}$, which lies outside this range. Here $h(0.4) = (0.4-\frac{1}{2})\log k+O(1)\to-\infty$ and $h(y_{\min})\to-\frac{3}{2}\log 2$. Since $h$ is convex, $\Phi\leq\max h+\frac{1}{2}\log\frac{3}{2}+o(1)<0$, and Proposition 25 shows these splits are safe at $\delta_+$. In both cases $L(p,q)>\frac{\pi}{2}+\delta_+>L_{\text{bal}}$. □

**Setting for near-balanced splits.** For the rest of this section $a,b>0$ are real, $N=a+b$, $x=a/N\in[0.4,0.6]$, and $\ell\in\Lambda := [\frac{\pi}{2}-\frac{1}{10},\frac{\pi}{2}+\frac{1}{10}]$. (For the splits of $k$ we have $N=2k-2$.) Let $\mu=\mu_{a,\ell}$ be the probability measure on $(0,\ell)$ with density $A_\ell/M_{a,b}(\ell)$, where $A_\ell(u) = \sin^a u\,\sin^b(\ell-u) = e^{-V(u)}$. Write $E$, Var, Cov for its moments, $u_1$ for its mode ($V'(u_1)=0$), and $\psi_\ell(u) = \log\sin u-\log\sin(\ell-u)$, so that $\partial_a V = -\psi_\ell$ when $b=N-a$ with $N$ fixed, and $\partial_b V = -\log\sin(\ell-u)$ when $a$ is fixed. Put $\Theta(u) = 1+|\log\sin u|+|\log\sin(\ell-u)|+|\cot u|+|\cot(\ell-u)|$, $B=[u_1-\frac{1}{4},u_1+\frac{1}{4}]$ and $B_1=[u_1-\frac{1}{14},u_1+\frac{1}{14}]$. Constants $C,C_m,c_0$ and the implied constants in $O(\cdot)$ do not depend on $a,\ell,N$. All statements hold for $N\geq N_1$, where $N_1$ may depend on the number of moments used.

**Lemma 29** (geometry)**.** *(i)* $u_1,\ \ell-u_1\in[0.58,0.99]$*. (ii)* $V''\ge N$ *on* $(0,\ell)$*. (iii) On* $B$*:* $\sin u,\sin(\ell-u)\ge 0.32$*,* $V''\le 10N$*,* $|V^{(j)}|\le CN$ *and* $|\psi_\ell^{(j)}|\le C$ *for* $j\le 5$*. (iv) On* $(0,\ell)$*:* $|\psi_\ell^{(j)}|\le C\Theta^{j+1}$ *and* $|V^{(j)}|\le CN\Theta^j$ *for* $1\le j\le 5$*.*

*Proof.* $u_1$ solves $a\cot u = b\cot(\ell-u)$, and $0<\partial_\ell u_1<1$ (proof of Lemma 27(a)). At $\ell=\frac{\pi}{2}$, $\tan^2 u_1 = a/b\in[\frac{2}{3},\frac{3}{2}]$, so $u_1$ and $\frac{\pi}{2}-u_1$ lie in $[0.684,0.887]$; this gives (i). (ii) holds because $\csc^2\ge 1$. On $B$ both $u$ and $\ell-u$ lie in $[0.33,1.24]$, which gives (iii). (iv) holds because the derivatives of $\log\sin$ are polynomials in cot. □

**Lemma 30** (tails and moments)**.** *(i)* $Z:=\int_0^\ell e^{-(V-V(u_1))}du\ge c_0N^{-1/2}$*, so the density of* $\mu$ *is at most* $C\sqrt{N}$*. (ii)* $E|u-u_1|^m\le C_mN^{-m/2}$*. (iii)* $E[\Theta^m;\,u\notin B_1]\le C_me^{-N/700}$*.*

*Proof.* (i) On $B$, $V-V(u_1)\le 5N(u-u_1)^2$. (ii) By Lemma 29(ii), $V(u)-V(u_1)\ge\frac{N}{2}(u-u_1)^2$, so $E|u-u_1|^m\le Z^{-1}\int|y|^me^{-Ny^2/2}dy$. (iii) Off $B_1$ we have $V-V(u_1)\ge N/392$. With $\theta=1/600$, $e^{-(V-V(u_1))}\le e^{-(1-\theta)N/392}e^{\theta V(u_1)}A_\ell^\theta$, and $e^{\theta V(u_1)}\le(\sin 0.58)^{-\theta N}\le e^{0.61\theta N}$. Hence the left side is at most $Z^{-1}e^{-0.00153N}\int_0^\ell\Theta^mA_\ell^\theta\,du$, and the last integral is bounded once $\theta a,\theta b\ge m+2$. □

**Lemma 31** (Stein identities)**.** *(i) If* $g\in C^1(0,\ell)$ *and* $|g|+|g'|\le C\Theta^m$*, then* $E\,g'(u)=E[g(u)V'(u)]$*. (ii) Let* $|c-u_1|\le C/N$ *and* $y=u-c$*. Then* $Ey=O(N^{-1})$*,* $Ey^2=V''(c)^{-1}+O(N^{-2})$*,* $Ey^3=O(N^{-2})$*, and* $E|y|^m=O(N^{-m/2})$*.*

*Proof.* (i) Integrate by parts on $(\epsilon,\ell-\epsilon)$; the boundary terms vanish because $e^{-V}$ vanishes to order $\ge 0.4N$ at 0 and $\ell$. (ii) The moment bound follows from Lemma 30(ii). By Lemma 29, $|V'(c)|=|V'(c)-V'(u_1)|\le C$, and on $B_1$ $V'(u)=V'(c)+V''(c)y+\frac{1}{2}V'''(c)y^2+R$ with $|R|\le CN|y|^3$. The contributions of $u\notin B_1$ to the identities below are $O(e^{-N/800})$ by Lemma 30(iii), since $|V'|\le N\Theta$. Applying (i) with $g=1$, $g=y^2$ and $g=y$: $0=V'(c)+V''(c)Ey+O(NEy^2)$, so $Ey=O(N^{-1})$. Next $2Ey=V'(c)Ey^2+V''(c)Ey^3+O(NEy^4)$, so $Ey^3=O(N^{-2})$. Finally $1=V'(c)Ey+V''(c)Ey^2+\frac{1}{2}V'''(c)Ey^3+O(NEy^4)$, so $Ey^2=V''(c)^{-1}+O(N^{-2})$. □

**Lemma 32** (smooth functionals)**.** *Let* $F,G\in C^3(0,\ell)$*, possibly depending on* $a$ *and* $\ell$*, with* $|F^{(j)}|,|G^{(j)}|\le C$ *on* $B$ *for* $j\le 3$ *and* $|F|,|G|\le C\Theta^m$ *on* $(0,\ell)$*. Then* $EF=F(u_1)+O(N^{-1})$*,* $\operatorname{Cov}(F,G)=F'(u_1)G'(u_1)/V''(u_1)+O(N^{-2})$ *(in particular* $\operatorname{Cov}(F,G)=O(N^{-1})$ *and* $\operatorname{Var}F=F'(u_1)^2/V''(u_1)+O(N^{-2})$*), and* $\kappa(F,G,G):=E[(F-EF)(G-EG)^2]=O(N^{-2})$*.*

*Proof.* Let $y=u-u_1$ and $Y=y-Ey$. By Lemma 31 with $c=u_1$: $Ey=O(N^{-1})$, $EY^3=Ey^3-3EyEy^2+2(Ey)^3=O(N^{-2})$, and $\operatorname{Var}y=V''(u_1)^{-1}+O(N^{-2})$. On $B_1$ write $F=F(u_1)+F'(u_1)y+R_F$ with $|R_F|\le Cy^2$, and $R_F=\frac{1}{2}F''(u_1)y^2+R_F'$ with $|R_F'|\le C|y|^3$; similarly for $G$. Contributions from $u\notin B_1$ are exponentially small by Lemma 30(iii). The first claim follows. Next, $\operatorname{Cov}(F,G)=F'G'\operatorname{Var}y+F'\operatorname{Cov}(y,R_G)+G'\operatorname{Cov}(y,R_F)+\operatorname{Cov}(R_F,R_G)$. Here $\operatorname{Cov}(y,R_F)=\frac{1}{2}F''\operatorname{Cov}(y,y^2)+\operatorname{Cov}(y,R_F')=O(N^{-2})$, since $\operatorname{Cov}(y,y^2)=Ey^3-EyEy^2$. The same holds for $G$, and $|\operatorname{Cov}(R_F,R_G)|\le(ER_F^2\,ER_G^2)^{1/2}=O(N^{-2})$. Finally $F-EF=F'(u_1)Y+Q_F$ with $|Q_F|\le C(y^2+N^{-1})$ on $B_1$, and similarly for $G$. Expanding, $\kappa=F'G'^2EY^3+E\,O(Y^2|Q|+|Y||Q|^2+|Q|^3)$, where $|Q|=|Q_F|+|Q_G|$, and this is $O(N^{-2})$. □

**Lemma 33** (odd functions)**.** *Let* $|c-u_1|\le C/N$*. Suppose that for* $|y|\le\frac{1}{15}$ *we have* $g(c+y)=-g(c-y)$ *and* $|g(c+y)|\le|y|^m$*, and that* $|g|\le C\Theta^{m'}$ *on* $(0,\ell)$*. Then* $Eg(u)=O(N^{-(m+1)/2})$*.*

*Proof.* Let $\Xi(y)=V(c+y)-V(c)$, so $Eg=\int g(c+y)e^{-\Xi(y)}dy/\int e^{-\Xi(y)}dy$. For $|y|\le\frac{1}{15}$ we have $c+y\in B_1$. There $|\Xi'(0)|=|V'(c)|\le C$, $N\le\Xi''\le 10N$ and $|\Xi'''|\le CN$, so $-C|y|+\frac{N}{2}y^2\le\Xi(\pm y)\le C|y|+5Ny^2$ and $|\Xi(y)-\Xi(-y)|\le C(|y|+N|y|^3)$. Hence $\int e^{-\Xi}\ge c_0N^{-1/2}$.

By oddness,

$$\Big|\int_{-1/15}^{1/15} g(c+y)e^{-\Xi(y)}dy\Big| = \Big|\int_0^{1/15} g(c+y)\big(e^{-\Xi(y)}-e^{-\Xi(-y)}\big)dy\Big|$$
$$\le C\int_0^\infty y^m(y+Ny^3)e^{Cy-Ny^2/2}dy = O(N^{-\frac{m+2}{2}}).$$

The rest of the domain contributes $O(e^{-N/800})\int e^{-\Xi}$: on $B_1$ because $\Xi(y)\ge -C|y|+\frac{N}{2}y^2$ with $|y|>\frac{1}{15}$, and off $B_1$ by Lemma 30(iii). □

**Lemma 34** (the angular mean)**.** *Let $\ell = L$ and let $s = s(a)\in(0,L)$ be defined by $E\sin(u-s)=0$, as in Theorem 3. Then $|s-u_1| = O(N^{-1})$, $s'(a) = \psi_L'(s)/V''(s)+O(N^{-2})$ (in particular $s' = O(N^{-1})$), and $s''(a) = O(N^{-2})$. Here derivatives are taken in $a$ with $b = N-a$ and $L$ fixed. Also $\partial_b s = O(N^{-1})$ when $a$ and $L$ are fixed.*

*Proof.* With $y_1 = u-u_1$, $s-u_1$ is an argument of $Ee^{iy_1}$, since $s = \arg Ee^{iu}$. Moreover $E\cos y_1\ge\frac12$, so $\operatorname{Re}Ee^{iy_1}>0$, and $|s-u_1|<L<\frac{3\pi}{2}$. Hence $s-u_1$ is the principal argument, $\tan(s-u_1) = E\sin y_1/E\cos y_1$, and $|s-u_1|\le C|E\sin y_1| = O(N^{-1})$ because $E\sin y_1 = Ey_1+O(E|y_1|^3)$. The $a$-derivative of the log-density of $\mu$ is $\psi-E\psi$, with $\psi=\psi_L$. Differentiating $E\sin(u-s(a))=0$ gives $s' = P/Q$ with $P=\operatorname{Cov}(\sin(u-s),\psi)$ and $Q = E\cos(u-s) = 1+O(N^{-1})$. Put $y = u-s$ and Taylor-expand $\psi$ at $s$ on $B_1$:

$$P = (\psi(s)-E\psi)E\sin y+\psi'(s)E[y\sin y]+\tfrac12\psi''(s)E[y^2\sin y]+O(Ey^4) = \psi'(s)Ey^2+O(N^{-2}).$$

This uses $E\sin y=0$, $E[y\sin y] = Ey^2+O(Ey^4)$ and $E[y^2\sin y] = Ey^3+O(E|y|^5) = O(N^{-2})$ (Lemma 31(ii) with $c=s$). Since $Ey^2 = V''(s)^{-1}+O(N^{-2})$, the formula for $s'$ follows. Next, $P' = \kappa(\sin(u-s),\psi,\psi)-s'\operatorname{Cov}(\cos(u-s),\psi) = O(N^{-2})$ and $Q' = \operatorname{Cov}(\cos(u-s),\psi)+s'E\sin(u-s) = O(N^{-1})$, both by Lemma 32. So $s'' = (P'Q-PQ')/Q^2 = O(N^{-2})$. Finally, differentiating $E\sin(u-s)=0$ in $b$ gives $\partial_b s = \operatorname{Cov}(\sin(u-s),\log\sin(L-u))/Q = O(N^{-1})$. □

**The three terms.** Fix $L=\frac{\pi}{2}+\delta\in\Lambda$ with $\delta\ge0$, and $L' = \pi-L$. Let $\mu=\mu_{a,L}$ (variable $u$) and $\mu'=\mu_{a,L'}$ (variable $v$), with expectations $E$ and $E'$. Theorem 3, together with the substitution $v=\pi-u$ in $I_{\text{out}}$, gives $I_{\text{in}} = \frac12 M_{a,b}(L)\,E|\sin(u-s)|$ and $I_{\text{out}} = M_{a,b}(L')\,E'\sin(v+s)$. Hence

$$G(a) := \log I_{\text{in}}-\log I_{\text{out}} = T_1+T_2-T_3-\log 2,$$
$$T_1 = \log\frac{M_{a,b}(L)}{M_{a,b}(L')},\qquad T_2 = \log E|\sin(u-s)|,\qquad T_3 = \log E'\sin(v+s),$$

with $b=N-a$ and $s=s(a)$. Primes denote derivatives in $a$.

**Proposition 35** (mass term)**.** $T_1''\le -4\delta/N+C(\delta^3N^{-1}+N^{-2})$.

*Proof.* $\partial_a^2\log M_{a,N-a}(\ell) = \operatorname{Var}_{\mu_{a,\ell}}(\psi_\ell)$. By Lemma 32 this equals $q(x,\ell)/N+O(N^{-2})$, where $q=\psi_\ell'(u_1)^2/K$ and $K = V''(u_1)/N = x\csc^2u_1+(1-x)\csc^2(\ell-u_1)$. Put $\gamma=\cot u_1$. The mode equation $x\cot u_1 = (1-x)\cot(\ell-u_1)$ gives $\psi_\ell'(u_1) = \gamma/(1-x)$, $K = 1+x\gamma^2/(1-x)$ and $q = \gamma^2/\big((1-x)(1-x+x\gamma^2)\big)$. Hence $\partial q/\partial\gamma = 2\gamma/(1-x+x\gamma^2)^2$, $\partial_\ell u_1 = (1-x)\csc^2(\ell-u_1)/K$ and $\partial_\ell\gamma = -\csc^2u_1\,\partial_\ell u_1$. At $\ell=\frac{\pi}{2}$ we have $\gamma^2=(1-x)/x$, $K=2$ and $\partial_\ell u_1=\frac12$, so

$$\partial_\ell q(x,\tfrac{\pi}{2}) = -\frac{\gamma}{2(1-x)^2}\cdot\frac{1}{2x} = -\frac{1}{4\,(x(1-x))^{3/2}}\le -2.$$

Since $q$ is real-analytic on $[0.4,0.6]\times\Lambda$, $q(x,L)-q(x,L') = 2\delta\,\partial_\ell q(x,\frac{\pi}{2})+O(\delta^3)$, and $T_1'' = (q(x,L)-q(x,L'))/N+O(N^{-2})$. □

**Proposition 36** (outer term)**.** $|T_3''|\le CN^{-2}$.

*Proof.* Let $F_a(v) = \sin(v + s(a))$ and $\chi = \psi_{L'}$. On the set $B$ of $\mu'$ we have $v + s \in [0.9, 2.24]$ by Lemmas 29(i) and 34, so $F_a \geq 0.78$ there and $E'F_a \geq \frac{1}{2}$. The $a$-derivative of the log-density of $\mu'$ is $\chi - E'\chi$. Therefore

$$(E'F_a)' = \mathrm{Cov}'(F_a, \chi) + s'E' \cos(v + s),$$
$$(E'F_a)'' = \kappa'(F_a, \chi, \chi) + 2s' \mathrm{Cov}'(\cos(v + s), \chi) + s''E' \cos(v + s) - s'^2 E'F_a.$$

By Lemmas 32 and 34, $(E'F_a)' = O(N^{-1})$ and $(E'F_a)'' = O(N^{-2})$, so $T_3'' = (E'F_a)''/E'F_a - ((E'F_a)'/E'F_a)^2 = O(N^{-2})$. □

**Proposition 37** (inner term)**.** $|T_2''| \leq CN^{-2}$.

*Proof.* We work in the frame that moves with $s$. Let $q_a(y) = A_L(s(a)+y)/M_{a,b}(L)$ for $y \in (-s, L-s)$, extended by 0; this is the density of $y = u - s$. Then $h(a) := E|\sin(u - s)| = \int |\sin y|\, q_a(y)\, dy$, and since $\partial_a V = -\psi$ and $\partial_a \log M_{a,b}(L) = E\psi$,

$$S_a(y) := \partial_a \log q_a(y) = \psi(s + y) - E\psi - s'V'(s + y),$$
$$\partial_a S_a(y) = 2s'\psi'(s + y) - \mathrm{Var}\, \psi - s''V'(s + y) - s'^2 V''(s + y).$$

$A_L$ vanishes to order $\geq 0.4N$ at 0 and $L$. Hence $q_a(y)$ and its first two $a$-derivatives (which equal $q_a S_a$ and $q_a(S_a^2 + \partial_a S_a)$ inside the support) extend continuously by 0. They are bounded uniformly on $\{|y| \leq L\} \times [a_0 - \eta, a_0 + \eta]$ for small $\eta > 0$, so we may differentiate under the integral sign. This gives $E_q[S_a] = 0$ and $E_q[S_a^2 + \partial_a S_a] = 0$ (differentiate $\int q_a = 1$ twice), together with

$$(\log h)'' = \mathrm{Var}_w(S_a) + E_w[\partial_a S_a], \qquad E_w[f] := E_q[|\sin y|\, f]/h.$$

*Size of $S_a$.* By Lemma 32, $E\psi = \psi(u_1) + O(N^{-1})$. By Lemma 34, $|s - u_1| = O(N^{-1})$, $|V'(s)| \leq C$ and $s' = \psi'(s)/V''(s) + O(N^{-2})$. Hence $S_a(0) = O(N^{-1})$, $\partial_y S_a(0) = \psi'(s) - s'V''(s) = O(N^{-1})$, and $\partial_y^2 S_a = \psi'' - s'V''' = O(1)$ on $B_1 - s$. So $|S_a(y)| \leq C(N^{-1} + y^2)$ for $s + y \in B_1$, while $|S_a| \leq CN\Theta(s + y)^2$ everywhere. *Lower bound for $h$.* The density of $u$ is at most $C\sqrt{N}$ (Lemma 30(i)). With $t = c_0 N^{-1/2}$ small enough we get $\mathbb{P}(|y| \leq t) \leq \frac{1}{2}$. Since $|y| < L < 1.7$ we have $|\sin y| \geq \frac{1}{2}\min(|y|, 1)$, hence $h \geq \frac{1}{2}E\min(|y|, 1) \geq \frac{t}{4} \geq cN^{-1/2}$. *Variance part.* $\mathrm{Var}_w(S_a) \leq E_q[|\sin y| S_a^2]/h \leq C\, E_q[|y|(N^{-1} + y^2)^2]/h + O(e^{-N/800}) = O(N^{-2})$. *Drift part.* Let $D(y) = \partial_a S_a(y) - \partial_a S_a(0)$. Since $E_w - E_q$ annihilates constants, $E_w[\partial_a S_a] = E_q[\partial_a S_a] + (E_w - E_q)[D]$, and $E_q[\partial_a S_a] = -E_q[S_a^2] = O(N^{-2})$. On $B_1 - s$, using $s' = O(N^{-1})$, $s'' = O(N^{-2})$ and Lemma 29(iii), we have $D(y) = \beta y + D_2(y)$ with $\beta = 2s'\psi''(s) - s''V''(s) - s'^2 V'''(s) = O(N^{-1})$ and $|D_2(y)| \leq Cy^2/N$. Everywhere $|D| \leq C\Theta(s + y)^4$. Moreover $E_q y = Eu - s = O(N^{-1})$, and $E_w y = E[|\sin(u - s)|(u - s)]/h = O(N^{-3/2})/h = O(N^{-1})$ by Lemma 33 with $m = 2$. Also $E_w y^2 + E_q y^2 = O(N^{-1})$. Hence $(E_w - E_q)[D] = \beta(E_w y - E_q y) + O(N^{-2}) = O(N^{-2})$. So $(\log h)'' = O(N^{-2})$, and $T_2 = \log h$. □

*Proof of Theorem 9.* Let $L = L_{\mathrm{bal}}$ and $\delta = L - \frac{\pi}{2}$. Then $\delta \geq \phi_k - \frac{\pi}{2} > \frac{\log k}{4k}$ for large $k$ by Theorem 5(i). The upper-bound part of the proof of Theorem 7 gives $\delta \leq \frac{\log k + \log 8\pi + 1}{4k} \leq \frac{2\log k}{k}$ for large $k$, so $L \in \Lambda$. By Lemma 28 it remains to treat splits with $0.41k < p \leq q$. For these $a = 2p - 1 \in [0.4N, 0.6N]$ when $k \geq 10$. By Propositions 35–37, for all real $a \in [0.4N, 0.6N]$,

$$G''(a) \leq -\frac{4\delta}{N} + C\Big(\frac{\delta^3}{N} + \frac{1}{N^2}\Big) \leq -\frac{\log k}{kN} + \frac{C'}{N^2} < 0 \qquad (k \geq K_B).$$

Moreover $G(a) = G(N - a)$. Indeed, $u \mapsto L - u$ and $s \mapsto L - s$ exchange $a$ and $b$ in $I_{\mathrm{in}}$, and $u \mapsto \pi + L - u$ does so in $I_{\mathrm{out}}$. The balanced exponents $a_{\mathrm{bal}} = 2\lfloor \frac{k}{2} \rfloor - 1$ and $N - a_{\mathrm{bal}}$ satisfy $G = 0$ at $L$. If $k$ is even, then $a_{\mathrm{bal}} = \frac{N}{2}$ is the maximum point of the strictly concave symmetric function $G$, so $G(a) < 0$ for every other $a$ in the window. If $k$ is odd, then $G(\frac{N}{2} \pm 1) = 0$ and strict concavity gives $G < 0$ outside $[\frac{N}{2} - 1, \frac{N}{2} + 1]$; every other admissible $a = 2p - 1$ satisfies

$|a - \frac{N}{2}| \geq 3$. In both cases every non-balanced split in the window has $I_{\text{out}} > I_{\text{in}}$ at $L$, i.e. $\Delta_{p,q}(L) > 0$, and hence $L(p,q) > L_{\text{bal}}$ by Theorem 3. Take $K_0 = \max(K_A, K_B, 10)$. □

**Monotonicity.**

**Proposition 38** (derivatives in $b$)**.** *Let $L = \frac{\pi}{2} + \delta \in \Lambda$ with $\delta \geq 0$, and let $T_1, T_2, T_3$ be as above, now regarded as functions of $b$ with $a$ and $L$ fixed. Then $\partial_b T_1 \geq 0.22\,\delta - C\delta/N$ and $|\partial_b T_2| + |\partial_b T_3| \leq C/N$.*

*Proof.* $T_1$. We have $\partial_b \log M_{a,b}(\ell) = m_\ell := E_\ell \log\sin(\ell - u)$, and $\partial_b T_1 = m_L - m_{L'}$. The log-density of $\mu_{a,b,\ell}$ has $\ell$-derivative $b\cot(\ell-u) - E\,b\cot(\ell-u)$, and $A_\ell$ vanishes to order $b$ at $u = \ell$. Hence $\partial_\ell m_\ell = E\cot(\ell-u) + b\,\mathrm{Cov}(\log\sin(\ell-u), \cot(\ell-u))$. By Lemma 32, and since $\partial_\ell u_1 = b\csc^2(\ell-u_1)/V''(u_1)$ (proof of Proposition 35),

$$\partial_\ell m_\ell = \cot(\ell-u_1) - \frac{b\cot(\ell-u_1)\csc^2(\ell-u_1)}{V''(u_1)} + O(N^{-1}) = \cot(\ell-u_1)\,(1-\partial_\ell u_1) + O(N^{-1}).$$

By Lemma 29(i), $\cot(\ell-u_1) \geq \cot 0.99 \geq 0.65$. Also $1 - \partial_\ell u_1 = x\csc^2 u_1/K$, and this is at least $0.4\csc^2(0.99)/\csc^2(0.58) \geq 0.17$. So $\partial_\ell m_\ell \geq 0.11 - C/N$ on $\Lambda$, and $m_L - m_{L'} \geq 2\delta(0.11 - C/N)$. $T_3$. The $b$-derivative of the log-density of $\mu'$ is $\log\sin(L'-v) - E'\log\sin(L'-v)$. So $\partial_b E'\sin(v+s) = \mathrm{Cov}'(\sin(v+s), \log\sin(L'-v)) + \partial_b s\,E'\cos(v+s) = O(N^{-1})$ by Lemmas 32 and 34, and $E'\sin(v+s) \geq \frac{1}{2}$. $T_2$. As in Proposition 37, $\partial_b \log h = E_w[S^b]$ with $S^b(y) = \log\sin(L-s-y) - E\log\sin(L-u) - \partial_b s\,V'(s+y)$. By Lemmas 32 and 34, $S^b(0) = O(N^{-1})$, $\partial_y S^b(0) = O(1)$ and $|\partial_y^2 S^b| \leq C$ on $B_1 - s$. Hence $E_w[S^b] = S^b(0) + \partial_y S^b(0)\,E_w y + O(E_w y^2) + O(e^{-N/800}) = O(N^{-1})$, because $E_w y = O(N^{-1})$ (Lemma 33) and $E_w y^2 = O(N^{-1})$. □

*Proof of Theorem 10.* Let $(p,q)$ be a split of $k$ with $0.42k \leq p \leq 0.58k$, and put $a = 2p-1$, $b_0 = 2q-1$, $L = L(p,q)$, $\delta = L - \frac{\pi}{2}$. Then $\delta \geq \phi_k - \frac{\pi}{2} > \frac{\log k}{4k}$ for large $k$ (Theorem 5(i)). Also $\delta \leq \frac{\log k}{k}$ for large $k$. Indeed, apply Lemma 27 to $(\min(p,q), \max(p,q))$, using $L(p,q) = L(q,p)$. At $\delta = \frac{\log k}{k}$ it gives $\Psi = 2r\log k + o(\log k)$, which exceeds $\log(2r_{ab}\sqrt{\pi N}) + o(1)$, so the split is unsafe there. For $b \in [b_0, b_0+2]$ we have $a/(a+b) \in [0.4, 0.6]$ once $k$ is large. Let $g(b) = \log I_{\text{in}} - \log I_{\text{out}}$, computed at $L$ with exponents $(a,b)$. Then $g(b_0) = 0$, and by Proposition 38,

$$g'(b) = \partial_b T_1 + \partial_b T_2 - \partial_b T_3 \geq 0.22\,\delta - \frac{C}{N} > \frac{0.05\log k}{k} - \frac{C}{2k-2} > 0 \qquad (k \geq K_2').$$

So $g(b_0+2) > 0$, i.e. $\Delta_{p,q+1}(L) < 0$, and $L(p,q+1) < L$ by Theorem 3. This proves endpoint monotonicity for $k \geq K_2'$. Now take $K_2 = \max(K_0, K_2')$. For $k \geq K_2$ we have $\phi_k = L(p_*, q_*)$ with $(p_*, q_*) = (\lfloor \frac{k}{2} \rfloor, \lceil \frac{k}{2} \rceil)$ (Theorem 9). Since $(p_*, q_*+1)$ is a split of $k+1$, $\phi_{k+1} \leq L(p_*, q_*+1) < L(p_*, q_*) = \phi_k$. □

*Numerical check of the three terms* (`numerics/bln_decomp.py`, `numerics/decomp.txt`). At $a = \frac{N}{2}$ and $L = L_{\text{bal}}$, as $N$ runs from 40 to 1280: $N^2 T_1''$ runs from $-11.7$ to $-19.3$, against $-4N\delta$ from $-12.1$ to $-19.4$; $N^2 T_2''$ runs from 0.06 to 0.005; $N^2 T_3'' \to -4$; and $N^2 G''$ runs from $-7.9$ to $-15.3$. These are consistent with the propositions above. For Proposition 38 (`numerics/bln_decomp_b.py`, `numerics/decomp_b.txt`), at the balanced split for $k = 20, \ldots, 640$: $N\partial_b T_1$ agrees with $2N\delta \cdot \frac{1}{2}\sqrt{x/(1-x)}$ to three decimals (from 2.98 to 4.84), $N\partial_b T_2 \to -\frac{1}{2}$, and $N\partial_b T_3 \to 0$. These numbers concern only near-balanced splits. They say nothing about the size of $K_0$ or $K_2$, which also depend on the $o(1)$ terms in Lemma 28, and we did not make the constants explicit.

## 8. Rigorous verification for $k \le 440$: proof of Theorem 8

Fix a split with $a = 2p-1$, $b = 2q-1$, and a rational $L \in (0,\pi)$. Expanding by the binomial theorem,

$$P(u) := \sin^a u \,\sin^b(u-L) = (2i)^{-a-b} \sum_{j=0}^{a}\sum_{l=0}^{b} \binom{a}{j}\binom{b}{l} (-1)^{a+b-j-l} e^{-i(2l-b)L}\, e^{i(2j-a+2l-b)u}.$$

Put $F_1(t) = \int_0^t P\sin$ and $F_2(t) = \int_0^t P\cos$. These are finite sums of $\int_0^t e^{inu}du = (e^{int}-1)/(in)$ with $n$ odd. In the notation of the proof of Theorem 3, $F = \cos s\, F_1 - \sin s\, F_2$. Since $P < 0$ on $(0,L)$, $s = \operatorname{atan2}(-F_1(L), -F_2(L))$, and

$$\Delta_{p,q}(L) = F(\pi) - F(s) = \cos s\,(F_1(\pi) - F_1(s)) - \sin s\,(F_2(\pi) - F_2(s)).$$

Every quantity here is evaluated in Arb ball arithmetic (python-flint, precision $200 + 6k$ bits, which absorbs the $2^{a+b}$ cancellation), giving a rigorous enclosure of $\Delta_{p,q}(L)$. By the sign pattern of Theorem 3, a certified $\Delta_{p,q}(L) > 0$ proves $L(p,q) > L$, and a certified $\Delta_{p,q}(L) < 0$ proves $L(p,q) < L$.

For each $k$ we take a rational $L^*$ (denominator $\le 10^{12}$) halfway between the floating-point values of $L_{\mathrm{bal}}$ and $\min_{p<\lfloor k/2\rfloor} L(p,k-p)$. We certify $\Delta_{\mathrm{bal}}(L^*) < 0$ and $\Delta_{p,k-p}(L^*) > 0$ for every other $p \le k/2$, which gives $L_{\mathrm{bal}} < L^* < L(p,k-p)$. The other comparisons in Theorem 8 use separating rational points in the same way. The enclosures of $\phi_k$ come from certified signs at rational points within $5\cdot 10^{-15}$ of $L_f \pm 10^{-12}$. The floating-point values only choose the test points; correctness rests on the certified signs. Code: `numerics/bln_rigorous.py`; the run for $k \le 200$ (all three claims) takes about 50 minutes on one core (output `numerics/rigorous_200.txt`), and the run `bln_rigorous.py 201 300` (the conjecture for $201 \le k \le 300$ and $\phi_{k+1} < \phi_k$ for $200 \le k < 300$) about 40 minutes (`numerics/rigorous_201_300.txt`). The range $301 \le k \le 400$ was run in four chunks, `bln_rigorous.py` with arguments 301 337, 338 363, 364 383 and 384 400, in parallel. Each chunk also certifies $\phi_{k+1} < \phi_k$ across its lower seam. This took about 85 minutes of wall time, at roughly 80 seconds per $k$ near $k = 350$ (`numerics/rigorous_301_400.txt`). Near the balanced split the quantities compared are of size about $2^{-(a+b)/2}$, so the rational test points are converted to balls at 4000 bits; with fewer bits the first $k$ of a run above $k \approx 370$ was not certified. The range $401 \le k \le 440$ (and $\phi_{k+1} < \phi_k$ for $400 \le k < 440$) was run as `bln_rigorous.py 401 440` on one core, at about 12 minutes per $k$ (`numerics/rigorous_401_440.txt`).

*Two further certificates.* Theorems 5(ii) and 6 use Theorem 8 for $k \le 300$. Above that, on a finite range, they use the explicit closed forms of Propositions 25 and 19. These are certified in Arb by `numerics/bln_lower_arb.py` (all 99977500 pairs $(k,p)$ with $301 \le k \le 2\cdot 10^4$; output `numerics/lower_arb.txt`) and by `numerics/bln_odd_upper_arb.py` (odd $301 \le k \le 200001$; output `numerics/odd_upper_arb.txt`). Both scripts read the enclosures of Theorem 8 from the output files above. `bln_rigorous.py` certifies $\phi_k \in (L_f - 1.01\cdot 10^{-12}, L_f + 1.01\cdot 10^{-12})$ and prints $L_f$ rounded to 12 decimals, so the scripts use the printed value $\pm 2\cdot 10^{-12}$.

## 9. Numerical evidence

**Computation of $L(p,q)$** (`numerics/bln_fast.py`). The quantities $s$, $I_{\mathrm{in}}$ and $I_{\mathrm{out}}$ of Theorem 3 are integrals of positive functions. We evaluate them in log scale with 1200-point Gauss–Legendre rules split at the peaks, and solve $\log I_{\mathrm{out}} = \log I_{\mathrm{in}}$ by Brent's method. Doubling the number of nodes changes $L$ by at most $6\cdot 10^{-14}$. The method reproduces $\phi_2 = 2\pi/3$, $\phi_3 = 1.962719002$ and $\phi_4 = 1.870658532$ to all digits.

**Results** (`numerics/sweep_300.txt`). For every $2 \le k \le 300$: the minimizing split is the balanced one (Conjecture 2, both parities); $p \mapsto L(p,k-p)$ is strictly decreasing on $1 \le p \le k/2$, with the

smallest gap $1.3 \cdot 10^{-7}$; and $\phi_k$ is strictly decreasing in $k$. Conjecture 11 is rigorously verified for $p+q<200$ (Theorem 8).

| $k$ | $\phi_k$ (numerical) | $4k(\phi_k - \frac{\pi}{2}) - \log k$ | lower (Prop. 25) | upper (Prop. 15) |
|---|---|---|---|---|
| 10 | 1.7115206092 | 3.3264 | +0.457 | 3.383 |
| 20 | 1.6493102586 | 3.2854 | +0.923 | 3.312 |
| 100 | 1.5904114962 | 3.2409 | +1.242 | 3.246 |
| 300 | 1.5782417442 | 3.2307 | +1.322 | 3.232 |
| $10^4$ | – | – | +1.382 | 3.2245 |

The last two columns are in the normalization $4k\delta - \log k$. The first gives the largest $c$ such that $\Phi < 0$ for all splits at $\delta = (\log k + c)/(4k)$. The second gives the smallest $\delta$ satisfying the sufficient inequality $2k \log\tan\alpha > \log(2kB(\frac{1}{2}, k)/c)$ in the proof of Theorem 4(i), for even $k$. The limit $\log(8\pi) = 3.22417$ is matched by the heuristic Laplace expansion of the criterion. For a split with $\rho = p/k$ and $r = 2\sqrt{\rho(1-\rho)}$, one has $I_{\text{in}} \approx M(L)/(2\sqrt{\pi N})$ ($N = 2k-2$), $I_{\text{out}} \approx rM(\pi - L)$, and $\log M(\frac{\pi}{2}+\delta) - \log M(\frac{\pi}{2}-\delta) \approx 4\sqrt{pq}\,\delta$. This predicts $L(p,q) \approx \frac{\pi}{2} + \frac{\log k + \log 8\pi + 2\log r}{4kr}$, which is minimized at $r = 1$ on the admissible range $r \geq 2/\sqrt{k}$. The predicted gaps $L(p, k-p) - L(\frac{k}{2}, \frac{k}{2}) \approx (\frac{\log k + \log 8\pi}{2} - 1)\frac{(\rho - 1/2)^2}{k}$ match the observed ones (for example $1.1 \cdot 10^{-7}$ predicted against $1.3 \cdot 10^{-7}$ observed at $k = 300$). The proof of Theorem 9 does not use this expansion. Instead it controls the second derivative of the criterion in the exponent (§7).

## References


[1] A. Barvinok, I. Novik, *A centrally symmetric version of the cyclic polytope*, Discrete Comput. Geom. 39 (2008) 76–99.

[2] A. Barvinok, S. J. Lee, I. Novik, *Neighborliness of the symmetric moment curve*, Mathematika 59 (2013) 223–249; arXiv:1104.5168.

[3] R. Sanyal, F. Sottile, B. Sturmfels, *Orbitopes*, Mathematika 57 (2011) 275–314.

[4] C. Vinzant, *Edges of the Barvinok–Novik orbitope*, Discrete Comput. Geom. 46 (2011) 479–487.

[5] J. G. Wendel, *Note on the gamma function*, Amer. Math. Monthly 55 (1948) 563–564.

[6] A. Prékopa, *On logarithmic concave measures and functions*, Acta Sci. Math. (Szeged) 34 (1973) 335–343.

[7] J. de Dios Pont, P. Ivanisvili, J. Madrid, *A new proof of the description of the convex hull of space curves with totally positive torsion*, Michigan Math. J. 76 (2026) 1043–1089.